

Science

Year 1

Student Manual



Science & Lab

The Hedge Bank Field Notebook

Science & Lab, Year 1, Student Book

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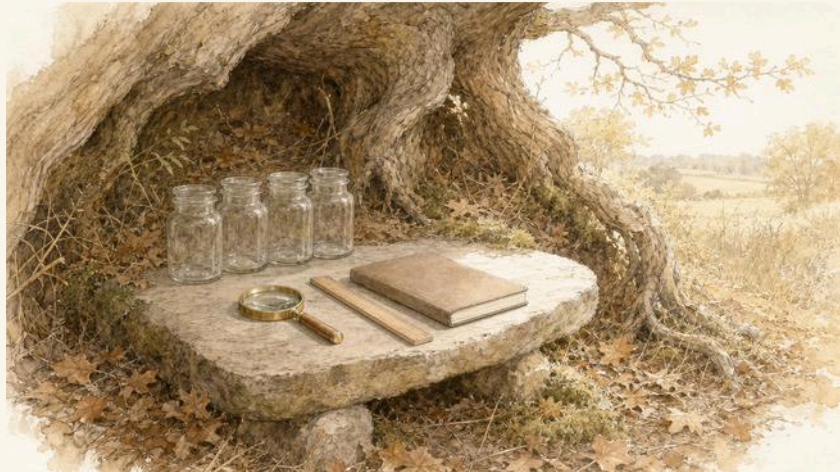
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WELCOME

Come out to the hedge bank



The Hollow, early on a bright morning, before anyone has arrived.

There is a bank at the edge of a field. Grass grows tall on it. Bramble and clover grow there too. Under the grass there are stones, and dry leaves, and crumbly brown earth.

All sorts of things live on that bank. Some of them are very small. Some of them are hard to see at all, until you look closely.

This book is about looking closely.

Six animals live on the hedge bank, and all six of them are curious. They want to know what things are made of, and why the Sun feels warm, and what makes a sound, and which things float. They ask questions, and then they find out.

You are going to do the same.

WELCOME

Six curious neighbours



Bram, Nib, Pod, Tor, Wren and Dart.

-  **Bram** the badger. He digs, so he knows what is under the grass. The brass hand lens is his.
-  **Nib** the wood mouse. His ears are enormous. He hears things nobody else notices.
-  **Pod** the snail. Pod is slow, and touches everything, so Pod knows how things feel.
-  **Tor** the toad. Tor sits very still on a warm stone and watches the sky all day.
-  **Wren**. She is the smallest, and the quickest, and she sings the loudest of them all.
-  **Dart** the frog. Dart pushes, pulls and jumps, and never stays still for long.

You will see all six of them again and again as you work through this book. Each one helps with the science they know best.

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HOW TO USE THIS BOOK

The parts of a page

Every topic in this book is built from the same few parts. Here is what each one looks like, so you know what to do when you see it.

Out on the bank

A question to talk about before you start. There is no wrong answer here.

☀ What Bram noticed

One of the six sees something, and wonders about it.

Then the topic begins.

✓ By the end you will

- know what this part of the book is for
- know what you will be able to do at the end

**WAKE UP
YOUR
SENSES**

A quick game to wake up your looking and listening.

Do you remember?

Something you already know, from a topic you have done before.

HOW TO USE THIS BOOK

Finding out, and having a go

🔍 Let us find out

The teaching part. Read it with your teacher, and look carefully at the picture beside it.

🔍 Watch me find out 1

How many legs has a woodlouse?

- 1 Look at the woodlouse through the hand lens.
- 2 Count the legs on one side. There are seven.
- 3 Count the legs on the other side. Seven again.

We found out: a woodlouse has fourteen legs, seven on each side

- 1 Now you have a go. Your turn items are numbered like this one.

Say your answer out loud first.

✂ The material you need

a hand lens, and a woodlouse from under a stone

⚠ Take care

Put every animal back exactly where you found it.

HOW TO USE THIS BOOK

Thinking, and checking

How do you know?

How do you know? This one asks you to say WHY you think so. It matters more than the answer.

♂ Find out together

Something to find out together, with a partner or in a group.

⌘ The material you need

the equipment you need, always listed beside the activity

*** Nib's notebook**

A surprising fact, from Nib.

→ Look a little closer

A harder one, if you finish early.

🏠 Hedge bank challenge

A puzzle for the whole group to argue about.

HOW TO USE THIS BOOK · CONTINUED

▣ WORDS WE USED



hand lens

makes small things
look bigger

□ Check what you can do

- 1 A question at the end of the unit.
- 2 If you can do these, you are ready for the next unit.

ALONG THE WAY

▣ Something to make

At the end of every unit there is something to make or to do, using what you have just learnt.

☐ I know what each part of a page is for.

	confident	almost there	needs more work
I can find my way around this book	☐	☐	☐

What a scientist needs



The kit you will use again and again this year.

You will not need very much. Most of the science in this book uses things you already have in the classroom, or things you can find outside.

GETTING SET UP • CONTINUED



hand lens

makes small things look bigger



jar

for holding what you find



ruler

for measuring how long



notebook

for drawing what you see



pencil

for writing and drawing



dish

for water, soil or seeds



trowel

for digging a little



magnet

for finding out what it pulls

& The material you need

a hand lens, a jar, a ruler, a notebook and a pencil



Take care

Wash your hands after you have been digging or holding animals.
Put every living thing back exactly where you found it.

How we find things out



Bram grew three bean shoots, and Nib wrote down how tall each one grew.

Scientists do not just look. They ask a question first, then they find out the answer, and then they say what they found.

The five steps we use every time

1

Ask a question

2

Say what you think will happen

3

Talk about what to do

4

Look carefully and record

5

Talk about what you found

You will use these five steps in every unit of this book. They are the same five steps every scientist in the world uses.

What we did

We put three pots on the windowsill and watered them the same.

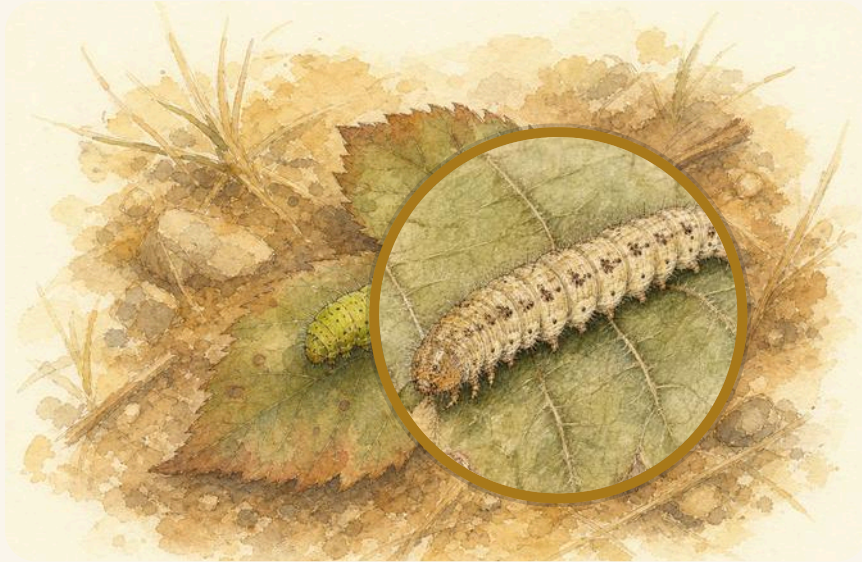
What we saw

.....
.....

Do not worry if what happens is not what you thought. That happens to scientists all the time. It is how they learn something new.

GETTING SET UP

Looking closely



A bramble leaf, and the same leaf through Bram's hand lens.

A hand lens makes small things look bigger. Hold it close to the thing you are looking at, then move it slowly up and down until you can see clearly.

GETTING SET UP · CONTINUED

Find out together

Look at your own thumb through the hand lens. Then look at a leaf. Then look at a grain of sand. Which one surprised you most?

The material you need

a hand lens, a bramble leaf and a few grains of sand

For teachers

The hand lens is the single most useful piece of equipment in this book, and pupils need practice with it before Unit 1. Let pupils hold the lens themselves rather than holding it for them: the skill is moving the lens, not looking through it.

Pupils at this age often try to look through the lens from far away. Model the movement once, close to the object, and let them find focus for themselves.

UNIT 1

Living things on the bank

Bram has found something moving under a
bramble leaf. Come and look closely with him.



Out on the bank

Look out of the window. Can you see something that is alive? Can you see something that has never been alive?

☀ What Bram noticed

I found a little green thing on a leaf, and it was moving. So I know it is alive. But how do I know the leaf is alive too?

Bram is right to wonder. Some living things move about, and some do not move at all. So moving cannot be the only test.

✓ By the end you will

- sort things into living and never alive
- name the parts of a plant
- find out what plants need to grow
- say how you know something is alive



Living and never-alive things lie side by side on the bank.



Alive, and never alive

1.1 Alive, and never alive

WAKE UP YOUR SENSES

Point to three things in the room that are alive. Now point to three things that have never been alive.

♀ Let us find out

A living thing **grows**. It **needs food** and **water** and **air**. It **makes young** ones like itself.

A snail is alive. A clover plant is alive too, even though it stays in one place. A flint pebble has never been alive. Nor has a glass jar.

Careful: a dry twig was **once** part of a living tree. It is not alive now, but it has not *never* been alive.

🔍 Watch me find out 1**Is this clover plant alive?**

- 1** It grows taller. That is one test.
- 2** It needs water. When Nib forgot to water it, it drooped.
- 3** It makes seeds, and new clover grows from them.

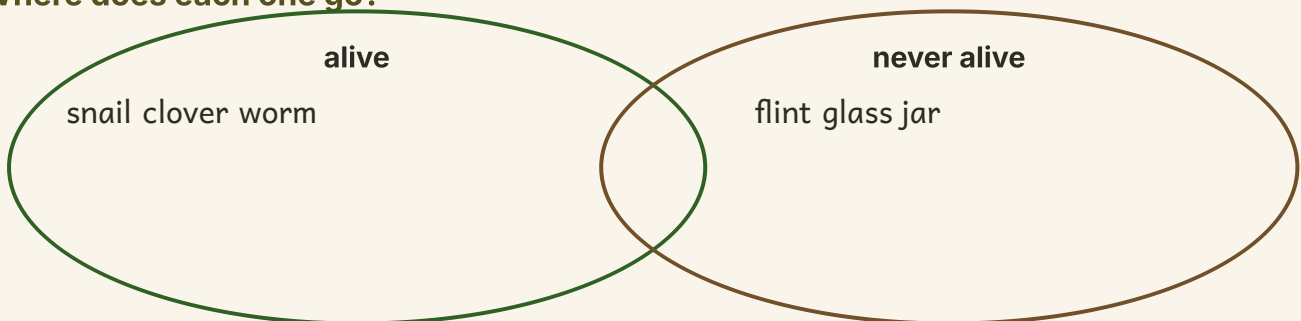
We found out: yes, the clover is alive, because it grows, needs water and makes new plants

How do you know?

Tor the toad sits on his stone for a whole hour without moving. Is Tor alive? How do you know?

Your turn

Where does each one go?



- 1 Draw a ring around the living things on this page.
- 2 Name one living thing you can see from your window.
- 3 Name one thing near you that has never been alive.
- 4 A snail moves. Does that prove it is alive? Say why.
- 5 A clover plant does not move about. Is it alive?
- 6 Write or draw one thing every living thing needs.

★ **Nib's notebook**

A snail can pull its whole body inside its shell, and close the door behind it with a plug of dried slime.

Your turn, and finding out together

A stone grows bigger every year.	TRUE / NOT TRUE
A clover plant needs water.	TRUE / NOT TRUE
A dry twig is alive.	TRUE / NOT TRUE
A woodlouse needs air.	TRUE / NOT TRUE

- 7 Tick TRUE or NOT TRUE for each sentence above.
- 8 Sort these into two groups: snail, pebble, moss, jar, worm.
- 9 Draw one living thing you found outside today.
- 10 Draw one thing outside that has never been alive.
- 11 Bram says the jar is alive because it holds water. Is he right?
- 12 Say two tests you can use to tell if something is alive.

The material you need

a hand lens, and a stone from outside with moss on it

Find out together

Turn over one flat stone outside. How many living things can you find under it? Look with the hand lens. Then put the stone back exactly as it was.

Take care

Put every animal back where you found it, and wash your hands afterwards.



The parts of a plant

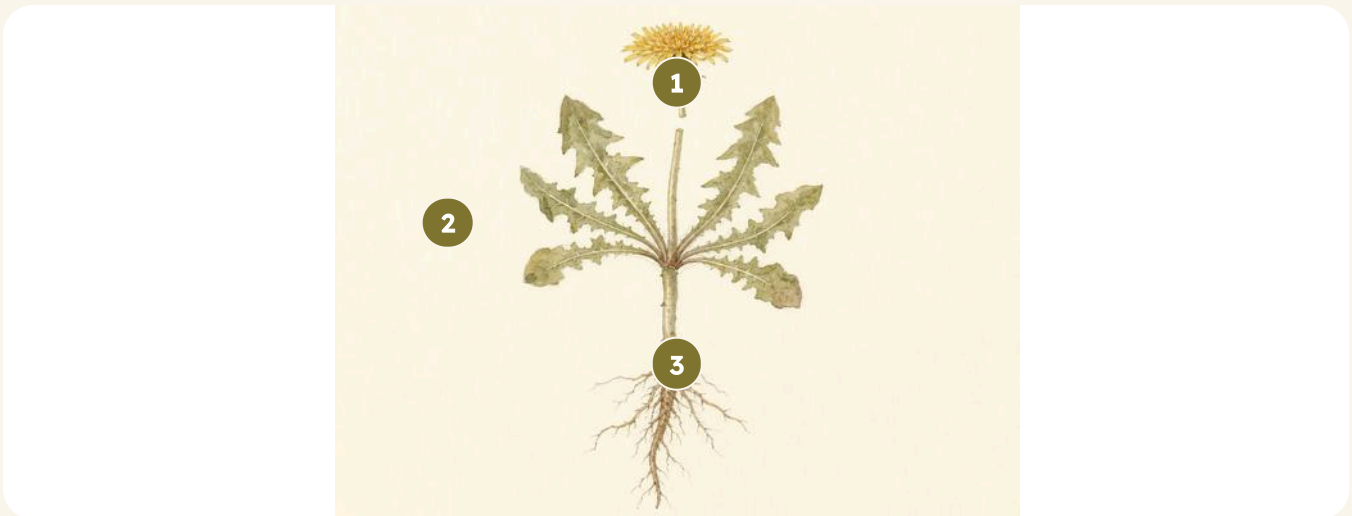
1.2 The parts of a plant

Do you remember?

You know that a clover plant is alive. What does it need so that it can grow?

♀ Let us find out

Every plant has the same few parts, and each part has a job to do. The **root** holds the plant in the soil, and drinks water. The **stem** holds the plant up. The **leaf** makes food for the plant out of light. The **flower** makes the seeds that new plants grow from.



- 1 flower
- 2 leaf
- 3 stem
- 4 root

This dandelion has 4 parts labelled. Find each one on a real plant outside.

Your turn

Watch me find out 2

Which part of this plant drinks the water?

- 1 Look at the part that is under the soil.
- 2 That part is the root.
- 3 The root takes in water from the soil.

We found out: the root drinks the water

1 Point to the root on the picture. Say what it does.

2 Point to the leaf. Say what it does.

3 Which part holds the plant up?

4 Which part makes the seeds?

5 Draw a plant. Label the root and the leaf.

6 Name a plant part you can eat.

How do you know?

Nib says the root is the most important part. Bram says the leaf is. What do you think? How do you know?

Your turn, and looking closely



Match each picture to its name: root, stem, leaf, flower.

7 Write the name under each picture above.

8 Sort the plant parts into two trays.

One tray for parts above the soil, one for parts below it.

9 Draw the leaf of a plant from outside. Draw its veins too.

10 How many leaves has your plant? Count them.

11 Does every plant have a flower on it today? Look and see.

12 Which part of a carrot do we eat: the root or the leaf?

& The material you need

a whole small weed pulled up gently, with its root still on, and a hand lens

♂ Find out together

Pull up one small weed with its root. Wash the soil off the root in a dish of water. Now look at the root with the hand lens. How many little roots come off the big one?

→ Look a little closer

Some plants keep their food in a fat root, so we can eat it. Name two roots you have eaten.



Plants and light

1.3 Plants and light

Do you remember?

You know what the leaf of a plant does. Say it again now.

♀ Let us find out

A plant cannot walk about to find its food. So it makes its own, and it makes it out of **light**.

The leaves do that job. A leaf is flat and wide so it can catch as much light as it can.

Soil is important too, but not as food. Soil holds the plant up, and holds the water the roots drink. A plant does not eat soil.

A plant kept in the dark grows tall, thin and pale, and then it dies. It is reaching for light it cannot find.



Two bean shoots, watered the same. One had light. One was kept under a box.

How do you know?

The shoot from under the box is pale and thin. How do you know it was the missing light that did that, and not the water?

Your turn

🔑 Watch me find out 3

Which shoot grew greener, the one in the light or the one in the box?

- 1 Both shoots got the same water, so water cannot be the reason.
- 2 Only one of them got light.
- 3 The one in the light is green and sturdy.

We found out: the shoot in the light grew greener, because a leaf needs light to make the plant its food

WHAT I THINK WILL HAPPEN*Draw or write what you think***WHAT REALLY HAPPENED***Draw or write what really happened*

Was what happened the same as what you thought would happen?

- 1** Draw the shoot that had light. Draw the one that did not.
- 2** Which of your two drawings looks healthier?
- 3** Say what a leaf uses light for.
- 4** Nib says the plant ate the soil. Is Nib right? Say why.
- 5** Where in your classroom would a plant grow best? Say why.
- 6** What would happen to grass under a big flat stone?

Your turn, and finding out together

A plant makes its own food.	TRUE / NOT TRUE
A plant eats the soil for its dinner.	TRUE / NOT TRUE
A leaf catches light.	TRUE / NOT TRUE
A plant grows well in a dark cupboard.	TRUE / NOT TRUE

- 7 Tick TRUE or NOT TRUE for the four sentences above.
- 8 Put the three shoots in order from palest to greenest.
- 9 Draw a leaf. Show how wide and flat it is.
- 10 Why is a leaf flat and not rolled up like a ball?
- 11 Name a place outside where grass will not grow. Say why.
- 12 What three things did both shoots get the same?

& The material you need

two small pots of soil, two bean seeds, water, and a box to keep the light out

♂ Find out together

Grow two bean seeds. Water them both the same. Put one on the windowsill and one under a box. Look at them every day for a week, and draw what you see.

*** Nib's notebook**

Some plants turn their leaves through the day to follow the Sun across the sky, so they catch light for as long as they can.



1.4 Plants need water

WAKE UP YOUR SENSES

Hold a leaf gently between your fingers. Is it stiff, or is it soft and floppy?

♀ Let us find out

Every living thing needs water, and a plant is no different.

The roots drink water out of the soil. The water travels up the stem to every leaf and flower.

When a plant has too little water it **wilts**: the stem goes soft and the leaves droop down. If it is watered soon enough it stands back up.



Two plants, side by side. One was watered. One was not.

What we did

We watered one plant and left the other one dry.

What we saw

Your turn

Watch me find out 4

One plant is drooping. What does it need?

- 1 The soil in its pot is pale and dry and cracked.
- 2 Its stem is soft and its leaves hang down. That is wilting.
- 3 Bram gave it water, and by the afternoon it stood up again.

We found out: the drooping plant needed water, and it stood up again once it had some

WHAT I THINK WILL HAPPEN

Draw or write what you think

WHAT REALLY HAPPENED

Draw or write what really happened

Was the wilted plant able to stand back up? Was that what you thought would happen?

- 1 Which part of a plant drinks the water?
- 2 Draw a plant that needs water. Show the leaves drooping.
- 3 How can you tell that soil is dry, just by looking?
- 4 Where does the water go after the root drinks it?
- 5 What does the word wilt mean? Say it in your own words.
- 6 Name one other living thing that needs water.

How do you know?

Wren says the dry plant is dead. Bram says it is only thirsty. How could you find out who is right?

Your turn, and finding out together

Tick the box that matches each plant

what we saw	the watered plant	the dry plant
stem stiff and upright	☐	☐
leaves drooping down	☐	☐
soil dark and damp	☐	☐
soil pale and cracked	☐	☐

- 7 Tick the right box on each row of the chart above.
- 8 Put these plants in order from driest to wettest.
- 9 How much water does your class plant need each week?
- 10 Draw the jug you would use to water it.
- 11 What would happen if you gave a plant no water for a month?
- 12 Say two things a plant needs, apart from water.

✂ The material you need

two potted plants the same, a jug of water, and a hand lens

♂ Find out together

Water one plant every day for a week. Give the other one no water at all. Draw both plants on the last day. Then water the dry one, and look again the next morning.

! Take care

Ask your teacher before you take any plant from outside.

Words we used in Unit 1

▣ WORDS WE USED



alive

it grows, feeds and makes young



never alive

it has never grown or fed



root

holds the plant, drinks water



stem

holds the plant up



leaf

makes food out of light



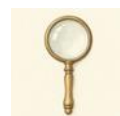
flower

makes the seeds



seed

a new plant grows from it



hand lens

makes small things look bigger

☐ Check what you can do

- 1** Name two tests that show something is alive.
- 2** Point to the root of a plant and say what it does.
- 3** Explain why a plant kept in a dark cupboard turns pale.
- 4** Say what happens to a plant with no water.
- 5** Name one thing on the bank that has never been alive.

Look what I can do!

- ☐ I can sort things into living and never alive.
- ☐ I can name the root, stem, leaf and flower of a plant.
- ☐ I can say that a leaf makes food out of light.
- ☐ I can say what a plant needs so that it can grow.
- ☐ I can say how I know something is alive.

	confident	almost there	needs more work
sorting living from never alive	☐	☐	☐
naming the parts of a plant	☐	☐	☐
saying what plants need	☐	☐	☐
saying how I know	☐	☐	☐



Bram and Nib look back at what they wrote down.

For teachers

Unit 1 establishes the two ideas the rest of the book leans on: that living things share a small set of needs, and that a claim in science carries a reason. Pupils who can answer the How do you know? prompts here will find Unit 5 straightforward.

The soil misconception is worth returning to. Many pupils will say a plant eats soil even after this unit, so ask the question again during Unit 4 when the class handles soil directly.

UNIT 2

Sounds all round the bank

Nib the wood mouse has heard something in
the dry leaves. Come and listen with him.



Out on the bank

Sit still and shut your eyes. How many different sounds can you hear? Which one is the loudest of them all?

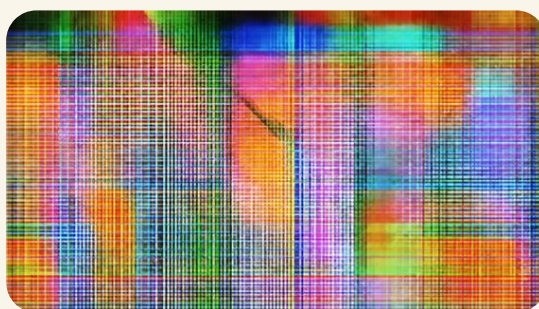
☼ What Nib noticed

I heard a rustle in the leaves, but I could not see anything at all. So how did the sound get all the way to my ear?

Nib has asked a good question. Sound is made a long way off, and then it travels through the air to us.

✓ By the end you will

- say how a sound is made
- name sounds that are loud and sounds that are quiet
- find out what sound can travel through
- say how you know what made a sound



The bank is never quiet. Every sound here has a maker.



2.1 Sounds all around us

WAKE UP YOUR SENSES

Hum with your hand on your throat. Can you feel the shaking under your fingers?

♀ Let us find out

Sounds are all around you. A sound is made when something **shakes**. The shaking is very fast, and it is often too small to see.

Pluck a tight string and you can watch it shake. Tap a drum and the skin shakes. Put one finger on the string and the shaking stops, so the sound stops too.

You hear sounds with your **ears**. You have two ears, and they help you tell which way a sound came from.

Watch me find out 1**What makes the sound when Nib plucks a string?**

- 1 Nib plucks a tight string. He can see it blur as it shakes.
- 2 He listens carefully. The string is making a sound.
- 3 He rests one finger on the string, and the shaking stops.

We found out: the sound stopped as soon as the shaking stopped, so the shaking is what makes the sound



A plucked string, a tapped drum, a shaken tin of rice and a blown reed. Every one of them is shaking.

How do you know?

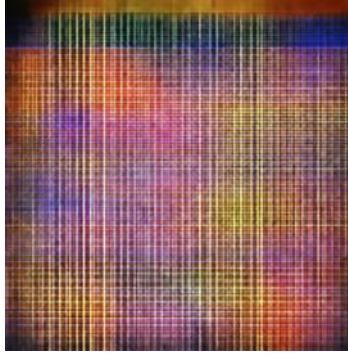
Nib taps a drum. The skin looks quite still to him. How do you know the drum skin is shaking?

Your turn

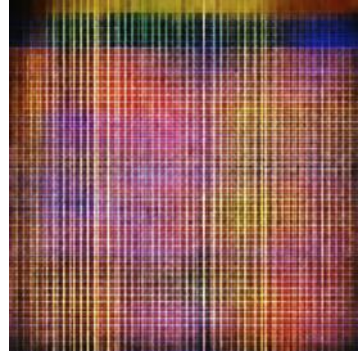
1



2



3



4



5



6



Six sound makers. Every one of them shakes when you play it.

1 Name each sound maker in the grid above.

2 Which one do you shake to play it?

3 Which ones do you blow to play them?

4 Which one do you tap to play it?

5 Draw one sound you heard on your way here today.

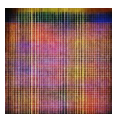
6 Which part of your body hears sounds?

★ **Nib's notebook**

A wood mouse has huge round ears for such a small head. Nib can hear an owl turn its head before he can see it.

Your turn, and finding out together

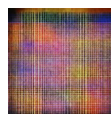
A sound is made when something shakes.	TRUE / NOT TRUE
You hear sounds with your eyes.	TRUE / NOT TRUE
A drum makes a sound when you tap it.	TRUE / NOT TRUE
A string that is quite still is making a sound.	TRUE / NOT TRUE



shaker



whistle



drum

- 7 Tick TRUE or NOT TRUE for each sentence above.
- 8 Sort these sound makers into a blow group and a tap group.
Use the bell, the drum, the whistle and the reed.
- 9 Draw a ring around the one you blow to play.
- 10 Write or draw two sounds that you like hearing.
- 11 Nib says a still string is making a sound. Is he right?
- 12 Say two ways to make a sound with a ruler.

& The material you need

a small tub with a lid, and a spoonful of dry seeds or tiny pebbles

♫ Find out together

Put the seeds in the tub and press the lid on. Shake the tub gently, then shake it hard. What is shaking inside to make the sound? Now try it with the lid off.

! Take care

Never put seeds, beads or pebbles in your ears, your nose or your mouth.



2.2 Loud and quiet

Do you remember?

You know that a shake makes a sound. What happens to the sound when the shaking stops?

♀ Let us find out

Some sounds are **loud**. Some sounds are **quiet**. A shout is loud. A whisper is quiet.

The same drum can make both. Tap it gently and the sound is quiet. Tap it harder and the sound is louder, because the skin shakes more.

A sound also gets quieter as you walk away from it. Stand close to the bell and it sounds louder. Stand far off and the same bell sounds quiet.



quietest

loudest

Five sounds in order, from quietest to loudest: a feather landing, a whisper, talking, a clap, a shout.

Watch me find out 2

Does a harder tap make a louder sound?

- 1 Nib sprinkles a few grains of sand on the drum skin.
- 2 He taps the drum gently. The grains hop a little way.
- 3 He taps the drum harder. The grains jump much higher.

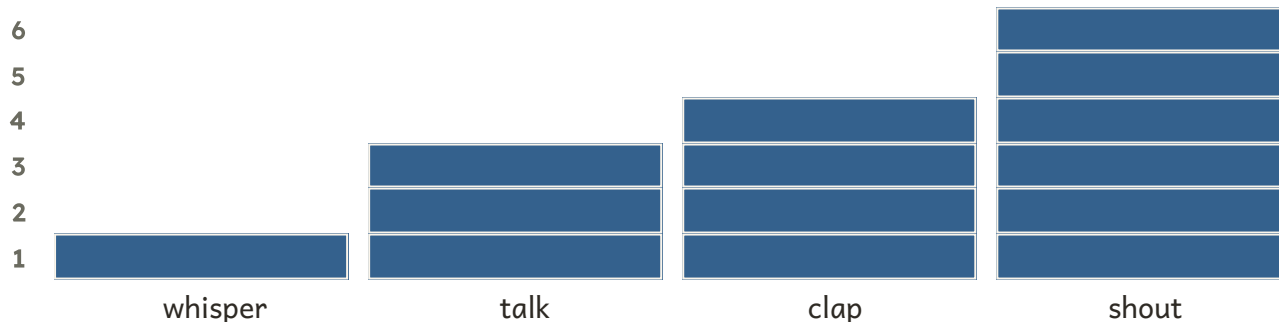
We found out: the harder tap made the grains jump higher, so a harder tap shakes the skin more and makes a louder sound

Your turn



Grains of sand jump on the drum skin as Nib taps it.

How loud? Nib counts it out in blocks



- 1 Which is louder, a whisper or a shout?
- 2 Name one loud sound you have heard today.
- 3 Name one quiet sound you have heard today.
- 4 Order these three sounds from quietest to loudest.
a clap, a whisper, a shout
- 5 Look at the block chart. Which sound has the tallest bar?
- 6 How many blocks tall is the bar for a clap?

How do you know?

Nib hears a bell but he cannot see it. He says the bell must be far away. How could he know that?

Your turn, and finding out together

The same bell, heard close up and heard far off.

WHAT I THINK WILL HAPPEN

Draw or write what you think

WHAT REALLY HAPPENED

Draw or write what really happened

Will the bell sound louder close up or ten steps away? Fill in the first box before you listen, then compare the two boxes.

- 7 Fill in the first box before anybody rings the bell.
- 8 Shake the bell close to your partner. Ask what they heard.
- 9 Now stand ten steps away and shake the bell again.
- 10 Compare what you thought with what really happened.
- 11 Tap a drum gently, then tap it harder. Which was louder?
- 12 Say why we whisper when a baby is asleep.

✂ The material you need

a hand bell, and a long clear space outside or in the hall

🔗 Find out together

One of you holds the bell and stands still. The other listens close by, then walks ten steps away and listens again. Swap over so you both hear it. Was the bell ever quite silent?

➔ Look a little closer

A shout carries a long way across a field, but you cannot shout for ever. How do people call to each other across water?



2.3 How sound travels to your ear

WAKE UP YOUR SENSES

Rest your ear flat on the table. Ask a friend to tap the far end softly.

♀ Let us find out

A sound has to travel to your ear. Most of the sounds you hear travel through the **air**, and you cannot see them going.

Sound can travel through other things too. It travels through **water**, through **wood** and along a tight **string**. Rest your ear on the table and ask a friend to tap the far end. The sound comes to you through the wood.

Sound always needs something to travel through. Air, water, wood and string all carry sound to your ear.

Through the air: the sound travels through the air to your ear.

Through the wood: the sound travels through the wood to your ear.



Nib holds his ear against the wood and listens.

Your turn

🔑 Watch me find out 3

Can a sound travel to Nib through wood?

- 1 Bram taps one end of a long plank, and Nib listens from the far end.
- 2 Nib lifts his head off the plank. The tapping sounds faint.
- 3 Nib rests his ear back on the wood. Now the tapping sounds clear.

We found out: the tapping sounded clearer through the wood, so sound can travel through wood as well as through air

Nib's listening test

we listened through	could we hear the tap?
the air	
the table top	
a tight string	
a sealed bag of water	

- 1 Tap the table while your ear rests on it.
- 2 Tick the chart above for each thing you tried.
- 3 Name two things that a sound can travel through.
- 4 Sound reached your ear through the wood. Draw its path.
- 5 Which was easier to hear through, the air or the wood?
- 6 Say why you cover your ears in a very loud room.

How do you know?

Nib puts his ear on the ground and hears Bram walking towards him. What is carrying the sound to Nib? How do you know?

Your turn, and looking closely

A string phone. The string must be pulled tight.

- 7 Make a string phone with a partner.
- 8 Pull the string tight. Now let it go loose.
- 9 Which worked better, a tight string or a loose one?
- 10 Say what carried your voice along to your partner.
- 11 Draw your string phone. Label the string and the cup.
- 12 Nib says his voice went through the air only. Is he right?

The material you need

two clean paper cups, a long piece of string, and a sharp pencil for the adult to use

Find out together

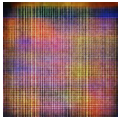
An adult makes one small hole in the bottom of each cup. Thread the string through and knot each end inside. Now walk apart until the string is tight. One of you whispers into a cup while the other listens.

Hedge bank challenge

Try the string phone again with the string resting on a table. Does your voice still get through? Say why you think that happened.

Words we used in Unit 2

▣ WORDS WE USED



sound

what we hear with our ears



loud

a big sound



quiet

a small sound



hollow

empty inside, so it can ring



vibrate

to shake to and fro very fast



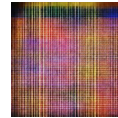
ear

the part we hear with



far away

a long way from you



listen

to use your ears on purpose

▣ Check what you can do

- 1 Name three things on the bank that make a sound.
- 2 Say what happens to a sound as you walk away from it.
- 3 Explain how a sound gets from a drum to your ear.
- 4 Name a sound that is too loud to be safe.
- 5 Say which part of your body you listen with.

Look what I can do!

- ☐ I can name things that make a sound.
- ☐ I can tell a loud sound from a quiet one.
- ☐ I can put sounds in order from quietest to loudest.
- ☐ I can say that sound needs something to travel through.
- ☐ I can say how I know a sound is coming from far away.

	confident	almost there	needs more work
naming what makes a sound	☐	☐	☐
telling loud from quiet	☐	☐	☐
saying how sound travels	☐	☐	☐
saying how I know	☐	☐	☐



Nib listens, with both ears turned forward.

For teachers

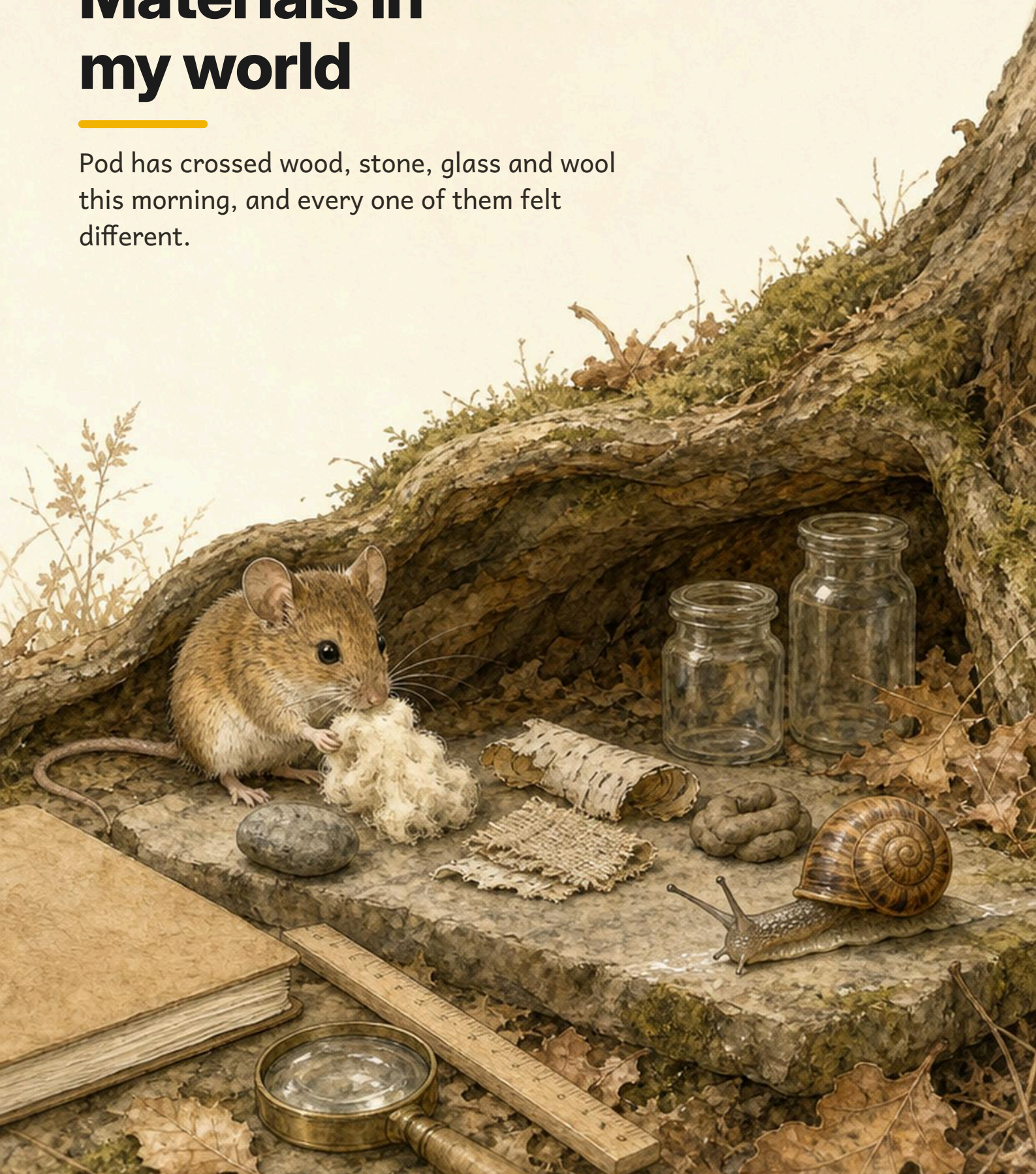
Unit 2 is the unit where pupils most often arrive with a wrong idea already formed: that sound travels through nothing at all, as it appears to in films set in space. The string telephone activity is the cheapest way to make the medium visible, so it is worth doing even if time is short.

Pupils at this age describe volume in terms of distance and distance in terms of volume, and the two need separating. Ask often whether a sound is quiet because it is small or quiet because it is far away.

UNIT 3

Materials in my world

Pod has crossed wood, stone, glass and wool
this morning, and every one of them felt
different.



Out on the bank

Touch the table. Now touch your jumper. Now touch the window.
Do they all feel the same?

☀ What Pod noticed

I go slowly, so I feel everything I cross. The stone was cold and hard. The wool was warm and soft. Why are they not the same?

Pod has noticed something important. Things are made of different stuff, and different stuff feels different.

✓ By the end you will

- name the materials things are made from
- say what a material is like
- sort materials into groups
- change the shape of a material



Wood, stone, glass, wool, cloth, bark and clay, side by side.



3.1 So many materials

WAKE UP YOUR SENSES

Look at your shoes. How many different materials can you find on one shoe?

♀ Let us find out

Everything around you is made of something. That something is called a **material**.

A jar is made of **glass**. A peg is made of **wood**. A jumper is made of **wool**. A spoon is made of **metal**.

One thing can be made of more than one material. A pencil is wood on the outside, with something soft and grey inside it.

And one material can be made into many different things. Wood becomes a peg, a ruler, a chair and a boat.



Nine materials. Can you name each one?

How do you know?

Pod says the window and the jar are made of the same material. How could you tell whether Pod is right?

Your turn

🔑 Watch me find out 1

What is this peg made of?

- 1 It is pale brown, and you can see lines running along it.
- 2 Those lines are the grain, and only one material has grain.
- 3 It came from a tree.

We found out: the peg is made of wood, and you can tell by its grain

- 1 Name the material each thing is made of.
- 2 Find something made of wood in this room.
- 3 Find something made of metal in this room.
- 4 What is a window made of?
- 5 Name two things made of wool.
- 6 Name one thing made of more than one material.

*** Nib's notebook**

Glass is made by heating sand until it melts. The window you look through was once a heap of sand on a beach.

Your turn, and finding out together



Write the name of the material under each one.

7 Write a material name under each of the eight pictures.

8 Sort the eight things into two trays.

One tray for things that came from a plant or animal.

9 Draw one thing made of glass.

10 Why is a boat not made of paper?

11 Name three things a piece of wood could be made into.

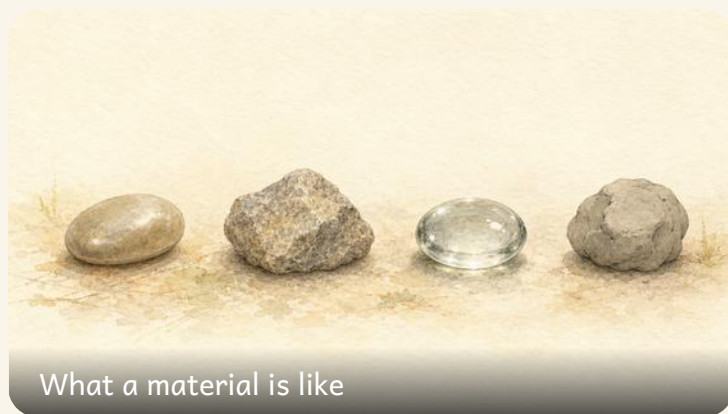
12 What is your chair made of? Look underneath as well.

& The material you need

a tray of small objects: a peg, a jar, a spoon, a stone, a scrap of cloth and a piece of paper

♣ Find out together

Take one object each from the tray. Say its material out loud. Then swap with a partner and do it again.



What a material is like

3.2 What a material is like

Do you remember?

You can name the material a thing is made of. Name the material of one thing near you now.

♀ Let us find out

We choose a material because of what it is **like**. What a material is like is called its **properties**.

Some materials are **hard** and some are **soft**. Some are **shiny** and some are **dull**. Some are **rough** and some are **smooth**. Some **bend** easily and some do not bend at all.

A window is made of glass because glass lets the light through. A boot is made of rubber because rubber keeps the water out. Nobody makes a boot out of paper.



A spoon and a glass pebble catch the light. Clay and bark do not.

Tick every box that is true for each material

material	shiny	dull	rough	smooth
metal spoon	☐	☐	☐	☐
grey clay	☐	☐	☐	☐
glass pebble	☐	☐	☐	☐
tree bark	☐	☐	☐	☐

Your turn

Watch me find out 2

Why is a raincoat made of plastic and not of paper?

- 1 Bram poured water on a square of plastic. It ran off in beads.
- 2 He poured water on a square of paper. It soaked in and went dark.
- 3 A coat has to keep the rain out.

We found out: plastic is waterproof and paper is not, so a raincoat is made of plastic



Water beads on the rubber. It soaks into the paper.

- 1 Name something shiny in this room.
- 2 Name something rough you can touch right now.
- 3 Is your jumper hard or soft?
- 4 Which bends more easily, a ruler or a rubber band?
- 5 Why are windows made of glass and not wood?
- 6 Name a material that keeps water out.

Your turn, and finding out together

WHAT I THINK WILL HAPPEN

Draw or write what you think

WHAT REALLY HAPPENED

Draw or write what really happened

Was the material you chose the best one? Was that what you thought would happen?

- 7 Put these four things in order from softest to hardest.
- 8 Rub each material. Which one feels roughest of them all?
- 9 Draw something made of a soft material.
- 10 Why is a saucepan handle not made of metal?
- 11 Choose the best material for a bucket. Say why you chose it.
- 12 Name two properties of wool.

How do you know?

Nib chose foil for a bucket because water runs off it. Wren chose plastic. Whose bucket would last longer, and how do you know?

& The material you need

squares of paper, cloth, plastic and foil, a dish of water, and a small spoon

Find out together

Put a spoonful of water on each square. Which squares keep the water out? Which let it through? Draw what you saw.

! Take care

Wipe up spilt water at once, so nobody slips.

→ Look a little closer

A tea towel soaks water up. A raincoat keeps it out. Both are cloth. How?



Sorting materials

3.3 Sorting materials

WAKE UP
YOUR
SENSES

Sort the class into two groups, using one rule. Can the others guess your rule?

♀ Let us find out

When you **sort** things, you put the ones that are alike together.

You can sort the same heap of things in more than one way. You could put all the wooden ones together. Or you could put all the shiny ones together instead, and then the wood and the metal would be mixed up.

So before you sort, you have to choose your **rule**. Then every single thing goes in the group its rule says.

One heap, sorted two ways at once

	shiny	dull
made of wood		
not made of wood		

How do you know?

Pod sorted a glass jar into the shiny group. Bram sorted the same jar into the see-through group. Were they both right? How do you know?

Your turn**Watch me find out 3****Sort these four things by the rule: does it bend?**

- 1 A rubber band bends easily, so it goes in the bending group.
- 2 A woollen sock bends too, so it goes there as well.
- 3 A glass pebble and a stone do not bend at all.

We found out: two things bend and two do not, so the rule splits them two and two

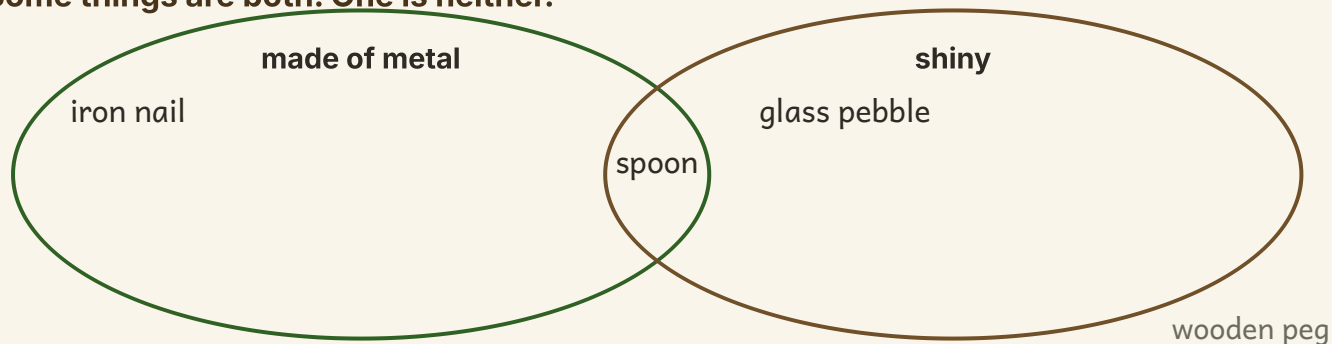
it bends

it does not bend

- 1** Write each thing into the right tray above.
- 2** Say your sorting rule out loud before you start.
- 3** Sort the buttons by colour.
- 4** Now sort the same buttons by size instead.
- 5** Did the same buttons end up together both times?
- 6** Name a rule you could use to sort shoes.

Your turn, and finding out together

Some things are both. One is neither.



- 7 Write each of the four things into the right place above.
- 8 Which thing goes in the middle? Say why.
- 9 Which thing goes outside both rings?
- 10 Sort your tray of objects from softest to hardest.
- 11 Make up a sorting rule of your own. Sort the tray by it.
- 12 Ask a partner to guess your rule.

& The material you need

a tray of about twelve small objects of different materials, and three shallow sorting trays

♣ Find out together

Sort the tray with a partner. One of you chooses the rule in secret and sorts. The other has to work out the rule.

*** Nib's notebook**

Everything that gets recycled has to be sorted first, and huge magnets do some of that sorting on their own.



Changing materials

3.4 Changing materials

Do you remember?

You know how to sort by a rule. What rule did you use yesterday?

♀ Let us find out

You can **change** a material without turning it into something else.

You can **squash** it, **bend** it, **twist** it or **stretch** it. Clay squashes flat. A twig bends. Wool stretches long.

Some materials spring back to how they were. A rubber band stretches and then goes back. That is called being **elastic**.

Other materials stay how you left them. Clay stays squashed. It is not elastic.



Stretched, bent, squashed and twisted. Four ways to change a material.

Four ways to change a material

squash it	bend it	twist it	stretch it		
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Your turn

Watch me find out 4

Which springs back, the rubber band or the clay?

- 1 Nib stretched the rubber band long, then let it go. It sprang back.
- 2 He squashed the clay flat, then took his thumb away.
- 3 The clay stayed flat. It did not spring back at all.

We found out: the rubber band sprang back because it is elastic, and the clay stayed squashed

WHAT I THINK WILL HAPPEN

Draw or write what you think

WHAT REALLY HAPPENED

Draw or write what really happened

Did the material go back to how it was? Was that what you thought?

- 1 Squash a ball of clay. Draw it before and after.
- 2 Bend a hazel twig gently. Does it spring back?
- 3 Stretch a rubber band. Now let it go. What happened?
- 4 Twist a piece of cloth. Does it stay twisted?
- 5 Name a material that springs back.
- 6 Name a material that stays how you left it.

Your turn, and finding out together

Clay springs back after you squash it.	TRUE / NOT TRUE
A rubber band is elastic.	TRUE / NOT TRUE
You can bend a glass pebble with your fingers.	TRUE / NOT TRUE
Wool can be stretched longer.	TRUE / NOT TRUE

7 Tick TRUE or NOT TRUE for the four sentences above.

8 Sort your materials from least bendy to most bendy.

9 Make a clay snake. Now make it into a ball again.

10 Is the clay still clay after you change its shape?

11 Which is easier to bend, a twig or a ruler?

12 Say what the word elastic means.

How do you know?

Wren says the clay is ruined once you squash it. Pod says it is still clay. Which of them is right, and how do you know?

✂ The material you need

a lump of clay, a rubber band, a hazel twig, a woollen sock and a square of cloth

🔗 Find out together

Try to squash, bend, twist and stretch each of the five things. Which one could you change in the most ways?

! Take care

Bend twigs gently, and keep them away from your face.

🔨 Hedge bank challenge

Bram says that once you have squashed clay flat you can never get the ball back. Is Bram right? Try it, then say what you found.

Words we used in Unit 3

▣ WORDS WE USED

**wood**

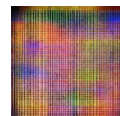
it comes from a tree
and has grain

**glass**

hard, smooth and see-
through

**wool**

soft and warm, from a
sheep

**metal**

hard and often shiny

**plastic**

made by people, keeps
water out

**clay**

soft sticky earth that
you can shape

**elastic**

it springs back to its
own shape

**material**

the stuff a thing is
made of

☐ Check what you can do

- 1** Name the material three things in this room are made from.
- 2** Say two properties of wool.
- 3** Explain what a sorting rule is.
- 4** Name a material that springs back after you stretch it.
- 5** Say why a window is made of glass.

Look what I can do!

- ☐ I can name the material a thing is made from.
- ☐ I can say what a material is like.
- ☐ I can sort materials using my own rule.
- ☐ I can squash, bend, twist and stretch a material.
- ☐ I can say which materials spring back.

	confident	almost there	needs more work
naming materials	☐	☐	☐
saying what a material is like	☐	☐	☐
sorting by a rule	☐	☐	☐
changing the shape of a material	☐	☐	☐



Pod's notebook, with a pressed leaf and a scrap of wool in it.

For teachers

Unit 3 is the unit where pupils learn that a property is a reason. Pressing for the word because after every material choice does more for their scientific language than any amount of naming practice.

Sorting the same tray twice by two different rules is the key activity. Pupils who see that the groups change when the rule changes are ready for the Carroll and Venn work in Unit 5.

UNIT 4

The Earth under our feet

Tor the toad has been sitting on the same stone all morning. Something has changed.



Out on the bank

Kneel down and touch the ground. Is it soft soil or hard stone?
Now look up. What can you see in the sky?

☀ What Tor noticed

I sat on my stone all morning. At first the stone was cold. By dinner time it was warm. Nobody lit a fire, so what warmed my stone?

Tor has found a good question. In this unit you will find out where that warmth comes from, and what the land under him is made of.

✓ By the end you will

- name the land, the water and the sky
- say what the Sun gives us, and stay safe in it
- sort rocks by how they look and feel
- say what soil is made of



Land, water and sky meet along the hedge bank.



Land, water and sky

4.1 Land, water and sky

WAKE UP
YOUR
SENSES

Stamp on the land. Splash in the water. Wave at the sky. Now say the three words in order.

♀ Let us find out

Kneel down and touch the ground. The ground under your feet is the **land**. Some land is soft soil, and some land is hard rock.

A puddle, a stream, a pond and the sea are all **water**. Above your head is the **sky**, full of air and clouds.

By day the **Sun** is in the sky. At night you may see the **Moon** and the stars. The Moon does not make its own light. It shines because sunlight bounces off it. The Moon travels around the Earth, and it is not a planet.

Land, water and sky together make our home, the **Earth**. The Earth is shaped like a huge ball. The ground near you looks level because the ball is so very big.

🔍 Watch me find out 1

Is a cloud part of the land, the water or the sky?

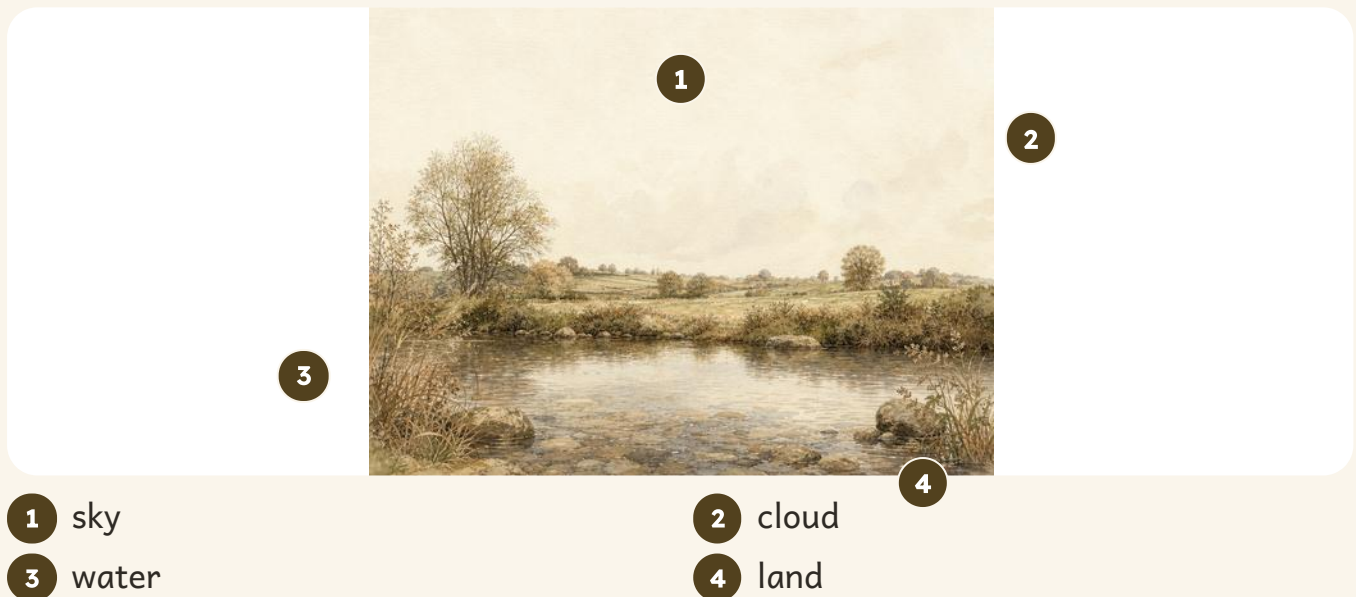
- 1 A cloud floats high above our heads.
- 2 Everything above our heads is the sky.
- 3 A cloud is made of millions of tiny water drops.

We found out: a cloud is in the sky, and it is made of water

How do you know?

Tor says there is water in the sky as well as in the pond. How do you know he is right?

Your turn



1 sky

3 water

2 cloud

4 land

This picture labels 4 things. Find every one of them outside today.

a fish

a cloud

a stone

a duck

- 1 Point to the land, the water and the sky in the picture.
- 2 Draw a ring above around the two things you find in water.
- 3 Name one thing you can see in the sky today.
- 4 Name one place near you where you can see water.
- 5 Is the ground outside your door soil or stone?
Go and look, then come back and write your answer.
- 6 Draw the land and the sky. Colour the sky.

* Nib's notebook

A cloud is made of millions of drops of water, each one far too small to see. A cloud is really water floating in the sky.

Your turn, and finding out together

WHAT I THINK WILL HAPPEN

Draw the sky as you think it will look tonight

WHAT REALLY HAPPENED

Draw the sky as you really saw it

Do this at bedtime, then bring your two pictures to school.

- 7 Draw what you think the sky will look like tonight.
- 8 Tonight, look at the sky. Draw what you really saw.
- 9 Compare your two pictures. What was the same?
- 10 Name one thing we see in the sky only at night.
- 11 Is the Moon in the sky every night?
Look for it on three nights and keep a note.
- 12 Tell a friend three things you have learnt about the sky.

✂ The material you need

a pencil, a sheet of dark paper and a window you can see the sky from

🔗 Find out together

Stand at the window with your partner. Find the line where the land meets the sky. Draw that line right across your paper. Then add the water you can see, and one thing in the sky.

→ Look a little closer

The line where the land meets the sky has a name. It is called the horizon. Look for it from a high place.



4.2 Heat and light from the Sun

Do you remember?

You know that the Sun is in the sky by day. What does the Sun do for us while it is up there?

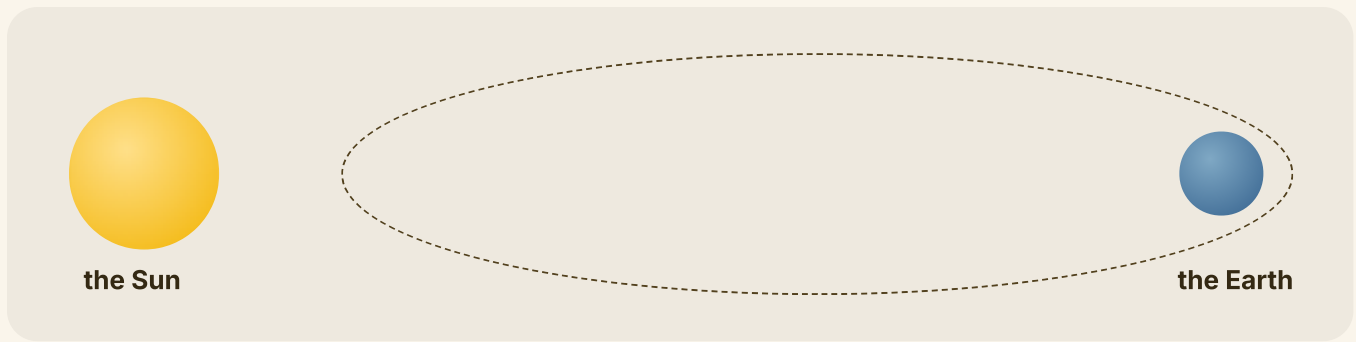
♀ Let us find out

The **Sun** is a star, and it is very far away. The Sun gives us **heat and light**.

The light from the Sun makes the day bright, so we can see. The heat from the Sun warms the land, the water and the air.

The Earth travels around the Sun. The Earth also spins. When your side of the Earth turns away from the Sun, it is night, and the ground cools down.

Sunlight cannot go through your body. That is why you have a **shadow**. A shadow is a dark patch where you block the light.



The Earth travels around the Sun. The Sun does not travel around the Earth.

! Take care

You must never look straight at the Sun, not even for a moment, and never through a hand lens or a window. The Sun can hurt your eyes.

🔍 Watch me find out 1

Which stone got warmer, the one in the sunshine or the one in the shade?

- 1** Tor put one stone in the sunshine and one in the shade.
- 2** He left both stones for a whole hour.
- 3** Then he felt each one with the back of his hand.

We found out: the stone in the sunshine was warmer, because heat and light come to us from the Sun

How do you know?

At night your playground is colder than it was at dinner time. How do you know the Sun makes that difference?

Your turn



The same bank in warm sunshine and in cool shade.

The Sun gives us heat and light.	TRUE / NOT TRUE
The Sun is a star.	TRUE / NOT TRUE
A shadow is a patch of light.	TRUE / NOT TRUE
The Earth travels around the Sun.	TRUE / NOT TRUE

1 Tick TRUE or NOT TRUE for each sentence above.

2 Name two things the Sun gives us.

3 Where does the heat that warms the playground come from?

4 Why do you have a shadow?

Talk it over with your partner before you write.

5 Order these three places from coldest to warmest: shade, indoors, sunshine.

6 Draw your own shadow on a sunny day.

★ Nib's notebook

Your shadow is longest early in the morning and late in the evening. It is shortest in the middle of the day.

Your turn, and finding out together

One child, one chalk mark, three times of day.

WHAT I THINK WILL HAPPEN

Draw where you think your shadow will be later

WHAT REALLY HAPPENED

Draw where your shadow really was

Do the shadow test in the morning, then again after dinner.

- 7** Draw where you think your shadow will be after dinner.
- 8** Go outside later. Draw round your shadow again.
- 9** Compare your mark with what you thought would happen.
- 10** Did your shadow move? Write yes or no.
- 11** Say one way people keep cool on a hot day.
- 12** Say one way the Sun helps a plant to grow.

& The material you need

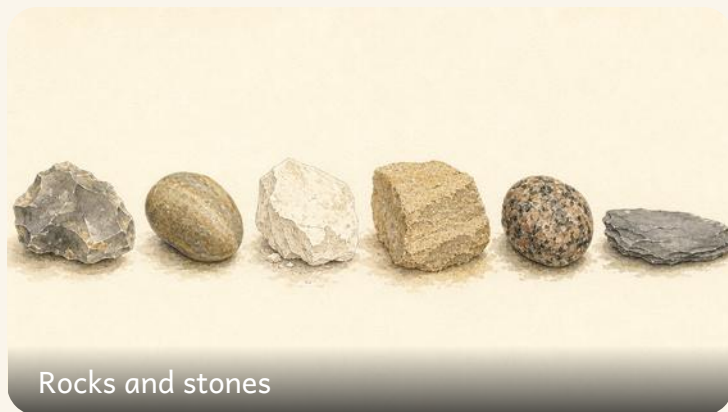
a sunny playground, a stick of chalk and a hat for each child

🔗 Find out together

Stand your partner in one spot in the playground. Draw round their shadow with chalk. Mark the spot so they can stand there again. Come back after dinner and draw round the shadow once more. What has changed?

! Take care

Wear a hat outside on a hot day. You must never look straight at the Sun.



4.3 Rocks and stones

WAKE UP YOUR SENSES

Hold a stone in your hand. Is it heavy or light, rough or smooth, warm or cold?

♀ Let us find out

Rock is hard. Under the soil, the whole land is made of rock. A small broken piece of rock is a **stone**.

Rocks are not all the same. **Flint** is very hard and shines where it is broken. **Chalk** is soft and white and rubs off on your hand.

Sandstone feels rough, like sugar. **Granite** is hard and speckled. **Slate** splits into thin grey sheets.

We build with rock because it is strong. A wall, a doorstep and a roof tile all began as rock in the ground.



Granite: hard, and speckled with grey, black and pink.

Watch me find out 1

Is this white stone chalk or flint?

- 1** Both chalk and flint can look white on the outside.
- 2** Tor rubbed the stone on his paper. It left a white mark.
- 3** Flint is very hard and leaves no mark at all.

We found out: the stone is chalk, because it is soft and rubs off on paper

How do you know?

Tor says his roof tile was once rock in the ground. How do you know he is right?

Your turn

1



2



3



4



5



6



Six rocks from the bank and the beach.

1 Write a name under each rock in the grid above.

The words are flint, pebble, chalk, sandstone, granite, slate.

2 Which rock is white and soft?

3 Which rock splits into thin sheets?

4 Find one stone outside. Draw it carefully.

5 Rub two stones together over paper. What came off?

6 Order three of your stones from smoothest to roughest.



Tor sorts his stones into trays on the wall.

*** Nib's notebook**

A hard flint often grows inside soft white chalk. When the chalk wears away, the flint is left behind on the beach.

Your turn, and looking closely



A broken flint through the hand lens. Look for the shine.

Test each stone

stone	hard	rough	shiny
flint	☐	☐	☐
chalk	☐	☐	☐
sandstone	☐	☐	☐

WHAT I THINK WILL HAPPEN

Draw what you think the lens will show

WHAT REALLY HAPPENED

Draw what the lens really showed

Look at your stone first with your eyes, then with the lens.

- 7 Tick the chart above for each stone you test.
- 8 Draw what you think you will see through the hand lens.
- 9 Look through the hand lens. Draw what you really saw.
- 10 Compare your two drawings. What did the lens show you?
- 11 Which of your stones would build the strongest wall?
- 12 Name two things near your school made of rock.

shiny stones

dull stones

✂ The material you need

six stones from outside, a hand lens, a dish of water and an old brush

♂ Find out together

Brush the soil off six stones in the dish of water. Look at each one through the hand lens. Sort them into the two trays above. Then tell your partner the rule you used.

 🪨 Hedge bank challenge

Find three stones of three different colours. Put them in order from lightest to darkest.





4.4 What soil is made of

Do you remember?

You know that rock lies under the land. What is the dark, crumbly stuff on top of it?

♀ Let us find out

Soil is the loose dark stuff on top of the land. Dig down with a trowel and you find stones, then solid rock.

Soil is made of tiny bits of broken rock mixed with bits of dead leaf. Water and air fill the little gaps in between.

Soil holds a plant up, and it holds water for the roots to drink. But a plant makes its own food out of light. To grow well a plant needs light, water, air and warmth.

Sandy soil feels gritty, and water runs straight through it. **Clay** soil feels smooth and sticky, and it holds the water.



1 soil

2 stones

3 rock

Watch me find out 1

What is soil made of?

- 1 Tor put a spoon of soil into a jar of water.
- 2 He shook the jar hard, then let it stand still.
- 3 Bits of dead leaf floated up. Grit and sand sank down.

We found out: soil is made of broken rock and bits of dead leaf, mixed together with water and air

How do you know?

Tor says there is air in the soil. How do you know he is right?

Your turn



Soil, sand, clay and a worm from the bank.



Soil through the hand lens: broken rock, bits of leaf, and gaps for air.

1 Write a name under each picture in the grid above.

2 Dig one small hole with the trowel. What did you find?

3 Name two things you can find in soil.

4 Which soil feels gritty, sandy soil or clay soil?

5 Which soil feels smooth and sticky?

6 Draw a worm. Say one way it helps the soil.

*** Nib's notebook**

One worm can pull a whole dead leaf down into its burrow. That is one way leaves turn into soil.

Your turn, and finding out together



A worm at work in its burrow under the bank.



The kit for the soil test.

WHAT I THINK WILL HAPPEN

Draw what you think will happen in the jar

WHAT REALLY HAPPENED

Draw what really happened in the jar

Shake the jar, then let it stand while you count to one hundred.

- 7** Draw what you think will happen to soil in water.
- 8** Shake the jar, then let it stand. Draw what really happened.
- 9** Compare what happened with what you thought.
- 10** Which sank to the bottom of the jar first?
- 11** Name one plant that grows in the soil near you.
- 12** Say three things a plant needs to grow well.

✂ The material you need

a trowel, a clear jar with a lid, a dish, some water and a spoon of soil

♂ Find out together

- 1 Half fill the jar with water.
- 2 Add one spoon of soil and screw the lid on tight.
- 3 Shake the jar, then stand it on the table.
- 4 Wait ten minutes, then draw what you can see.

! Take care

Wash your hands with soap and warm water after you touch soil.

Words we used in Unit 4

▣ WORDS WE USED



Earth

the planet we all live on



the Sun

it gives us heat and light



shadow

made where something blocks the light



rock

hard stone from the ground



soil

the loose dark earth plants grow in



sand

tiny loose grains of worn rock



clay

soft sticky earth that holds water



worm

it mixes the soil as it moves

☐ Check what you can do

- 1** Name the planet you live on.
- 2** Say two things the Sun gives us.
- 3** Explain how a shadow is made.
- 4** Name two things you would find in soil.
- 5** Say how a rock and a handful of soil are different.

Look what I can do!

- ☐ I can name the planet I live on.
- ☐ I can say that the Sun gives us heat and light.
- ☐ I know never to look straight at the Sun.
- ☐ I can say how a shadow is made.
- ☐ I can name what soil is made of.
- ☐ I can sort rocks by how they look and feel.

! Take care

Remember the most important rule in this unit: you must never look straight at the Sun, not even for a moment, and never through a hand lens.

	confident	almost there	needs more work
naming our planet	☐	☐	☐
saying what the Sun gives us	☐	☐	☐
explaining a shadow	☐	☐	☐
sorting rocks	☐	☐	☐
saying what is in soil	☐	☐	☐



Tor on his warm stone, with the shadow moving round it.

For teachers

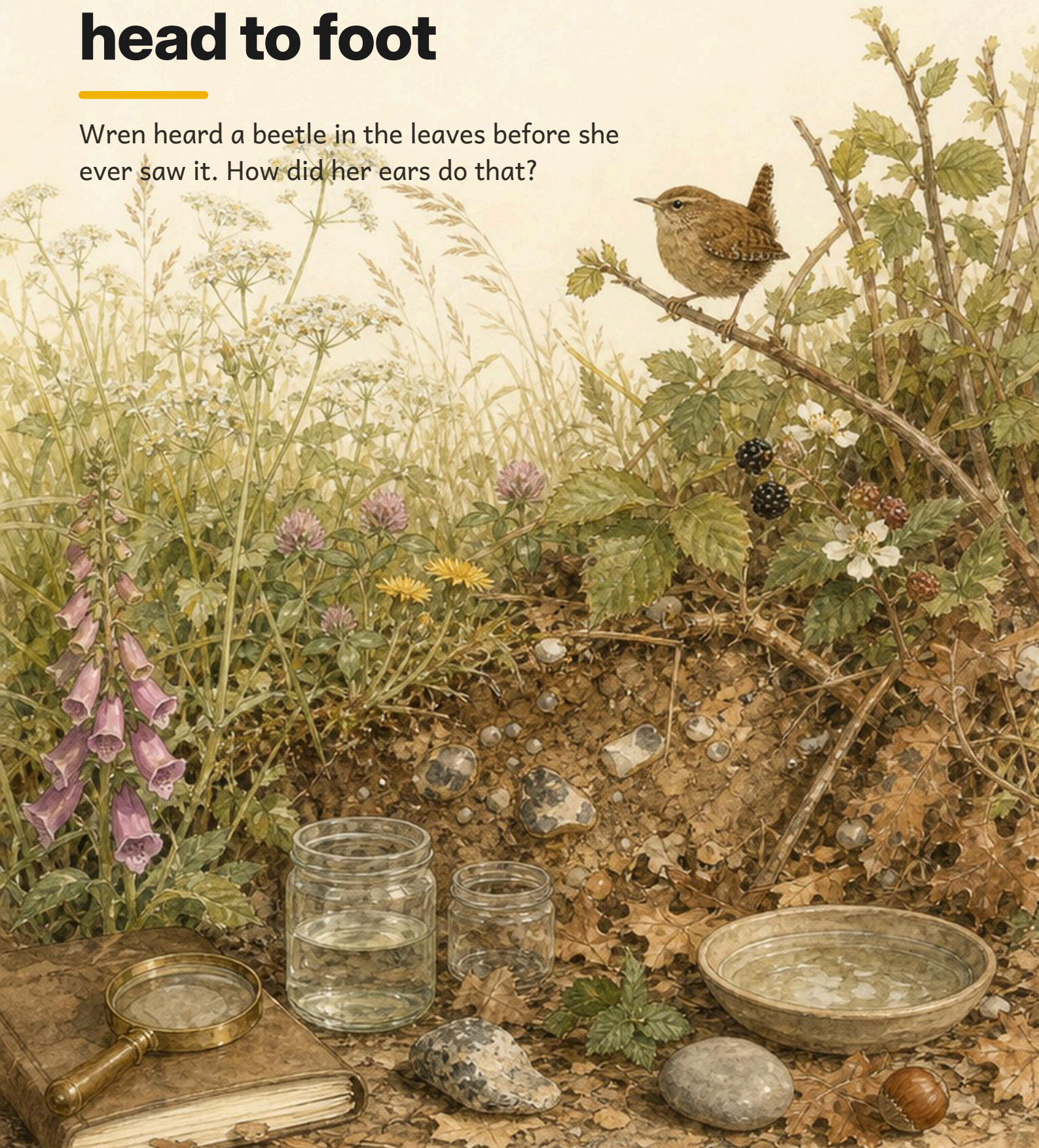
Unit 4 is the unit where the safety rule matters most. Say the Sun rule out loud at the start of every outdoor session in this unit, not just once when the page is read.

Pupils commonly believe soil is simply dirt. Handling it with a hand lens, and finding the sand, clay, dead leaves and living animals in it, is what changes that. Link it back to Unit 1: the plant does not eat the soil, it drinks the water the soil holds.

UNIT 5

Our bodies, head to foot

Wren heard a beetle in the leaves before she ever saw it. How did her ears do that?



Out on the bank

Stand up and wiggle every finger. How many can you count? Now close your eyes and keep still. What can you still hear?

☀ What Wren noticed

I heard a beetle under the dead leaves, but I could not see it at all. My ears found it before my eyes did.

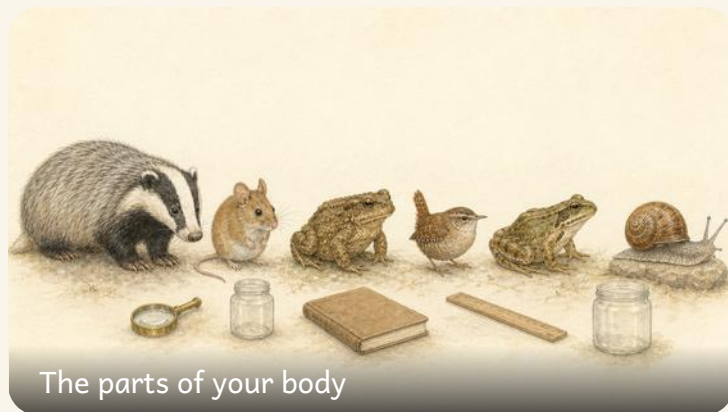
Wren has noticed something worth chasing. In this unit you will name the parts of your body, try out all five of your senses, and find out what keeps you well.

✓ By the end you will

- name the parts of your body
- name your five senses and what each one does
- say how people are the same and different
- say what keeps your body well



Wren and her friends, from head to foot.



5.1 The parts of your body

WAKE UP YOUR SENSES

Touch your head. Touch your knees. Touch your ears.
Now do all three as fast as you can.

♀ Let us find out

Your body is made of many parts, and every part has a job to do.

You have one **head**, two **arms** with a **hand** on each, two **legs** with a **foot** on each, and a **tummy** in the middle.

On your head are your **eyes**, your **ears**, your **nose** and your **mouth**. Your **hair** grows there too.

The parts work together. Your legs walk while your arms swing.
Deep inside your chest your **heart** beats all day and all night, and you never have to think about it.

🔍 Watch me find out 1

Which parts of Wren move when she jumps?

- 1 Wren watched herself jump in the long mirror.
- 2 Her legs pushed hard, and her feet left the floor.
- 3 Her arms swung up above her head at the same time.

We found out: her legs, her feet and her arms all move when she jumps

How do you know?

Wren says your heart is working this very minute. How do you know she is right?

Your turn



- | | | |
|---------|-------|--------|
| 1 head | 2 arm | 3 hand |
| 4 tummy | 5 leg | 6 foot |

This picture labels 6 parts. Point to every one of them on yourself.

head

arm

hand

tummy

leg

foot

1 Point to each part in the picture and say its name.

2 How many fingers do you have altogether?

Count them twice to be sure.

3 Draw yourself. Label your head, your arm and your leg.

4 Which part of you helps you hold a pencil?

5 Which part of you helps you kick a ball?

6 Label your drawing with the words from the word bank.

*** Nib's notebook**

There are twenty-six bones in each of your feet. More than a quarter of all your bones are down there in your feet.

Your turn, and finding out together

WHAT I THINK WILL HAPPEN

Draw or write what you think your heart will do

WHAT REALLY HAPPENED

Draw or write what your heart really did

Feel your heart first while you are sitting quite still.

7 Draw what you think your heart will do after ten hops.

8 Do ten hops. Now feel your heart again.

9 Compare what happened with what you thought.

10 Did your heart beat faster or slower after hopping?

11 Name two parts of you that moved when you hopped.

12 Name one part of you that never stops working.

& The material you need

a clear space to hop in, and a clock your teacher can time ten seconds on

♂ Find out together

Sit very still. Put two fingers flat on the side of your neck and find your heartbeat. Count the beats while your teacher times ten seconds. Now hop ten times and count again. Tell your partner what changed.

→ Look a little closer

Squeeze a soft ball once every second. That is roughly the work your heart does, all day and all night.



5.2 Our five senses

Do you remember?

You have named the parts of your body. Which parts are on your head, and what are they for?

♀ Let us find out

You find out about the world with your **five senses**.

You **see** with your eyes. You **hear** with your ears. You **smell** with your nose. You **taste** with your tongue. You feel by **touch**, with your skin, and best of all with your fingertips.

Very often you use more than one sense at a time. When you eat a blackberry you see it, you smell it and you taste it all at once.

Your senses keep you safe as well. Your skin tells you that water is too hot, and your ears tell you that a car is coming.



1 eyes

2 ears

3 nose

4 tongue

5 skin

Watch me find out 1

Which sense tells Wren that the leaf is mint?

- 1 Wren shut her eyes, so seeing could not help her.
- 2 She rubbed the leaf and held it near her nose.
- 3 The smell was sharp and sweet, and she knew it at once.

We found out: her sense of smell told her that the leaf was mint

How do you know?

Wren says she can find the bell with her eyes shut. How do you know she can?

Your turn

see



ears



hear



skin



smell



eyes



taste



tongue



touch



nose



A feather through the hand lens. Stroke it first, then look closely.

- 1 Match each sense to the part of you that uses it.
- 2 Which sense tells you a flower is purple?
- 3 Which sense tells you the bell is ringing?
- 4 Close your eyes. Which senses can you still use?
- 5 Name two senses you use when you eat an apple.
- 6 Which sense warns you that the water is too hot?



Lavender. Rub one leaf gently and smell your fingers.

**Take care**

Only smell the things your teacher hands to you. Never put anything in your mouth unless your teacher says that it is safe.

Your turn, and finding out together



Four smells from the hedge bank.



quiet

loud

WHAT I THINK WILL HAPPEN

Write the sound you think you will hear most

WHAT REALLY HAPPENED

Write the sound you really heard most

Shut your eyes for the sound test, and keep very still.

7 Write which sound you think you will hear most often.

8 Sit still for one minute. Tally every sound you hear.

9 Compare your tally with what you thought.

10 Order these three sounds from quietest to loudest.

A whisper, a hand clap, a school bell.

11 Which sense did you use to sort the sounds?

12 Draw the thing you heard most often.



Three sounds to name with your eyes shut.

& The material you need

a bell, a jug of water, an apple and a chair for each child

♫ Find out together

Your partner shuts their eyes. Ring the bell, pour the water, then tap the apple on the table. Can your partner name each sound in the right order? Now swap over.

★ Nib's notebook

Your nose helps your tongue. When a cold blocks your nose, your dinner seems to have almost no flavour at all.



The same, and different

5.3 The same, and different

WAKE UP YOUR SENSES

Stand beside a friend. Find three ways you are the same. Now find three ways you are different.

♀ Let us find out

We are all people, so we are alike in many ways. We all grow. We all have a heart that beats, and eyes to see with.

We are all different too. Your eyes may be brown, grey, blue or green. Your hair may be dark or fair, curly or straight. Your hands may be bigger or smaller than your partner hands.

Some things about you keep changing. You grow taller, your feet grow longer, and your first teeth fall out.

Looking for what is the **same** and what is **different** is a big part of science.

Watch me find out 1**Are Wren and Tor the same, or different?**

- 1** Both of them grow, and both of them have eyes.
- 2** Wren has feathers, and Tor has damp brown skin.
- 3** Wren has two legs, and Tor has four.

We found out: they are the same in some ways and different in others

How do you know?

Wren says no two children in your class are exactly alike. How do you know she is right?

Your turn

Four children, alike in some ways and not in others.

The eye colours in one class

What we found	Tally	How many
brown		7
blue		4
grey		3
green		1
altogether		15

There are **15** children in that class. Now make a tally for your own class.

brown

blue

grey

green

1 Write one way you and your partner are the same.

2 Write one way you and your partner are different.

3 What colour are your eyes? Draw them.

4 Count how many in your group have brown eyes.

5 Draw a ring around the commonest eye colour above.

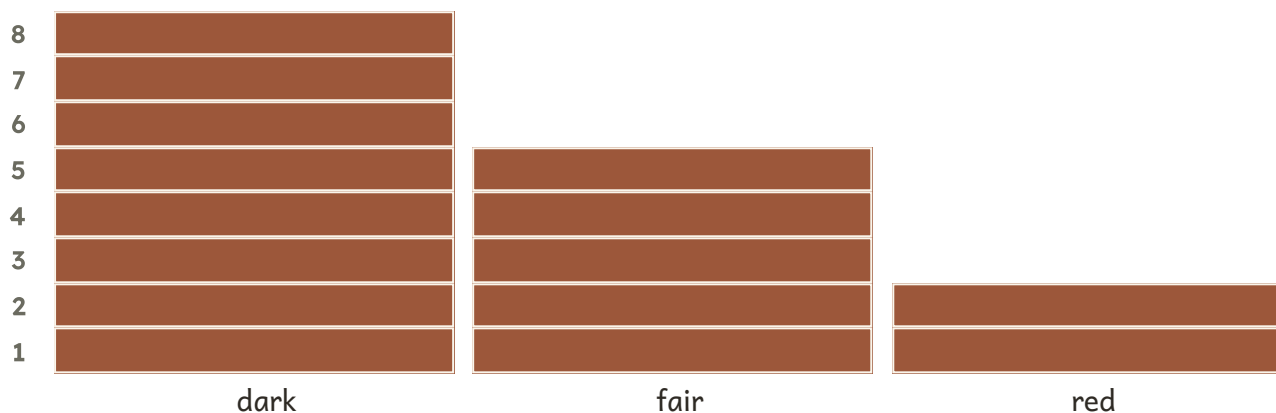
6 Is everybody in your class the same height?

★ Nib's notebook

The pattern of little lines on the tip of your finger is yours alone.
Nobody else in the world has the very same one.

Your turn, and finding out together

Hair colour in one class



WHAT I THINK WILL HAPPEN

Write how many you think are taller than you

WHAT REALLY HAPPENED

Write how many really are taller than you

Guess first, then line up and check.

7 Write how many children you think are taller than you.

8 Line up with three friends. Who is tallest?

9 Compare what you found with what you thought.

10 Order your group from shortest to tallest.

11 Sort your class into two trays: curly hair and straight hair.

12 Name one thing about you that has changed since you were small.

curly hair

straight hair

✂ The material you need

a wall to stand against, a long strip of paper for each child and a pencil

👫 Find out together

Stand up straight against the wall. Your partner holds a paper strip from the floor to the top of your head, and marks it with a pencil. Do this for four children. Then put the four strips in order from shortest to longest.

👯 Hedge bank challenge

Find two children with the same eye colour but different hair colour. Draw them both.



Keeping your body well

5.4 Keeping your body well

Do you remember?

Your body grows and works all day long. What do you think it needs from you in return?

♀ Let us find out

Your body needs a few simple things every day so that it can stay well and grow.

It needs **food** of many kinds, and plenty of **water**. It needs **sleep**, because you do most of your growing while you are asleep. It needs **exercise**, so run and climb and skip every day.

Keeping **clean** keeps germs away. Germs are far too small to see. Wash your hands with soap and warm water before you eat and after you go to the toilet, and brush your teeth twice a day.



Food, water, sleep, exercise and a good wash.

Watch me find out 1

Did the soap really make a difference?

- 1 Wren rubbed garden soil on both of her hands.
- 2 She washed one hand in water alone, and one with soap.
- 3 Then she dried each hand on a clean white towel.

We found out: the towel from the hand washed with soap was much cleaner

How do you know?

Wren says soap does a better job than water on its own. How do you know she is right?

Your turn

1 	2 	3 	4 
--	--	---	--

Four things that help to keep you well.

Your body grows most while you are asleep.	TRUE / NOT TRUE
Germes are big enough to see.	TRUE / NOT TRUE
Running makes your heart beat faster.	TRUE / NOT TRUE
Water alone cleans as well as soap.	TRUE / NOT TRUE

1 Tick TRUE or NOT TRUE for each sentence above.

2 Name three foods that help you grow well.

3 How many hours do you sleep each night?

4 Name one game that makes your heart beat faster.

5 Draw yourself brushing your teeth.

6 When should you wash your hands?

Try to think of two different times in the day.

* Nib's notebook

Germes are so small that a thousand of them could sit on the head of a pin without touching each other.

Your turn, and finding out together



Soap, warm water, and a good long rub.

WHAT I THINK WILL HAPPEN

Draw what you think the soap will take off

WHAT REALLY HAPPENED

Draw what the soap really took off

Look at the clean towel before you start, then again at the end.

- 7** Draw what you think the soap will take off your hands.
- 8** Wash your hands with soap and warm water.
- 9** Compare the towel with what you thought would happen.
- 10** How long did you rub your hands together?
- 11** Say why we wash our hands before we eat.
- 12** Draw a poster telling your class to wash their hands.

& The material you need

soap, a bowl of warm water, a clean towel, a sponge and a little garden soil

♣ Find out together

- 1 Rub a little soil on both of your hands.
- 2 Wet your hands in the bowl of warm water.
- 3 Rub soap all over them while you count to twenty.
- 4 Rinse them, then dry them on the clean towel.
- 5 Look at the towel. What can you see?

! Take care

Use warm water, never hot water. Wipe up any spills so that nobody slips.

Words we used in Unit 5

▣ WORDS WE USED



body

all of you, from head
to toe



senses

the five ways we find
things out



touch

what we feel with our
skin



smell

what we find out with
our nose



taste

what we find out with
our tongue



hearing

what we find out with
our ears



different

not the same as
another one



healthy

well, and looked after

▣ Check what you can do

- 1 Name four parts of your body.
- 2 Name the five senses.
- 3 Say which part of you each sense uses.
- 4 Name one way you are different from your partner.
- 5 Say two things that keep you well.

Look what I can do!

- ☐ I can name the parts of my body.
- ☐ I can name my five senses.
- ☐ I can say which part of me each sense uses.
- ☐ I can say how I am the same as my friends, and how I differ.
- ☐ I can say what keeps me well.
- ☐ I can wash my hands properly.

	confident	almost there	needs more work
naming body parts	☐	☐	☐
naming the five senses	☐	☐	☐
finding out by touching, smelling and listening	☐	☐	☐
saying how we are same and different	☐	☐	☐
saying what keeps me well	☐	☐	☐



Wren, the smallest of the six, and the loudest.

For teachers

Unit 5 needs handling with care in any class where a pupil has a difference in one of the senses, or a visible physical difference. The unit is built so that difference is normal and interesting rather than a deficit, and the same-and-different topic is the place to say so plainly.

The senses give the best reasoning practice in the book, because a pupil can be asked how they know with their eyes shut. Guessing games where sight is unavailable make the answer genuinely evidential rather than remembered.

UNIT 6

Pushes and pulls on the bank

Dart has given a little ball one push, and away it went down the bank. Come and watch it roll.



Out on the bank

Look around you. Can you see something that is moving? Can you see something that is staying still?

☼ What Dart noticed

I gave a little ball one push and it rolled all the way down the bank. Then it stopped by itself. Why did it stop?

Dart is right to wonder. A push starts a thing moving. Then something else has to slow it down and stop it again.

✓ By the end you will

- name the ways that things move
- tell a push from a pull
- make a ball roll faster and further
- test which things float and which sink
- find out what a magnet pulls



A ball rolls down the bank while Dart watches it go.



6.1 Things that move

WAKE UP YOUR SENSES

Roll a pencil across your table. Now slide your book across it. Which one went further?

♀ Let us find out

Things move in different ways. A ball can **roll**. A sledge can **slide**. A top can **spin**. A swing goes **to and fro**.

Some things move quickly and some move slowly. A thing that is still stays still until something gives it a **push** or a **pull**.

Careful: one thing can move in more than one way. A ball can roll, and it can slide along the wet grass too.

Watch me find out 1**How does this marble move across the floor?**

- 1** Nib gives the marble one gentle push with her paw.
- 2** The marble goes over and over on its round side.
- 3** It does not drag along. It keeps turning as it goes.

We found out: the marble rolls, because it is round all over

How do you know?

Dart says every round thing rolls. Is Dart right? How do you know?

Your turn

Rolling, sliding, spinning and going to and fro.

Tick every way each thing can move.

thing	it rolls	it slides	it spins
a ball	☐	☐	☐
a marble	☐	☐	☐
a spoon	☐	☐	☐
a coin	☐	☐	☐
a wooden peg	☐	☐	☐

1 Draw a ring around the things that can roll.

2 Name one thing in this room that can slide.

3 Which things in the chart can spin?

4 Draw a swing. Add an arrow to show which way it goes.

5 Write or draw one thing that is staying still.

6 Say two ways a ball can move.

★ **Nib's notebook**

A snail cannot roll or slide. It glides along on a slippery ribbon of slime that it lays down in front of itself.

Your turn, and finding out together

things that roll

things that do not roll

A ball can roll.	TRUE / NOT TRUE
A coin can spin.	TRUE / NOT TRUE
A still thing starts moving all by itself.	TRUE / NOT TRUE
A swing goes to and fro.	TRUE / NOT TRUE

7 Tick TRUE or NOT TRUE for each sentence above.

8 Sort these things into the two trays.

a ball, a marble, a coin, a wooden peg, a spoon, a stone

9 Put these in order from slowest to fastest.

a crawling snail, a walking child, a rolling ball

10 Roll the ball and the peg. Which one stopped first?

11 Dart says a spoon can roll. Test it and say what happened.

12 Draw one thing outside that was moving today.

& The material you need

a ball, a marble, a wooden peg, a spoon and a smooth floor

♂ Find out together

Give the ball one gentle push along the floor. Now give the wooden peg the very same push. Which one travels further? Take turns, and tell your group what you found.



6.2 Pushes and pulls

Do you remember?

You know that a still thing needs something to start it moving. What could start it moving?

♀ Let us find out

A **push** sends something away from you. A **pull** brings something nearer to you. A push and a pull are both **forces**.

You push a door shut. You pull a drawer open. You push a swing, and it goes away from your hands. Then you pull it back again.

Careful: you cannot see a push or a pull. You can only see what it does, so watch which way the thing moves.



The blue arrow shows a push. The red arrow shows a pull. Both of them are forces.

Your turn

Watch me find out 2

Is Dart pushing or pulling this little cart?

- 1 Dart has the handle of the cart in both hands.
- 2 He leans back, and the cart comes nearer to him.
- 3 It is moving towards Dart, not away from him.

We found out: Dart is pulling, because the cart comes nearer to him

Tick a push or a pull for each one.

what you do	a push	a pull
shut a door	☐	☐
open a drawer	☐	☐
kick a ball	☐	☐
tug a rope	☐	☐
press a bell	☐	☐
pull up a weed	☐	☐

- 1 Tick a push or a pull in the chart above.
- 2 Name one push you have done today.
- 3 Name one pull you have done today.
- 4 Does a pull bring a thing nearer or send it away?
- 5 Draw yourself pushing a swing. Add an arrow for the push.
- 6 Open your book. Was that a push or a pull?

How do you know?

Nib pushes a heavy log and it will not move. Did Nib use a force?
How do you know?

Your turn, and finding out together

push the boat



lift the bucket



pull the string

- 7 Draw a ring around the picture that shows a pull.
- 8 Push your chair in. Say which way it moved.
- 9 Sort these words into pushes and pulls.
shut, tug, press, drag, kick, reel in
- 10 Draw two arrows on the boat. Label the push and the pull.
- 11 A magnet brings a nail nearer. Is that a push or a pull?
- 12 Say one way a pull helps you every day.

The material you need

a bucket with a little water in it, a long piece of string and a smooth floor

Find out together

Tie the string to the bucket handle. Pull the bucket towards you along the floor. Now push it away with your hands. Which was easier? Talk about it with your partner.

→ Look a little closer

A pull can act at a distance. A magnet pulls a nail without touching it at all, and so does the Earth when it pulls a leaf down.



6.3 Making things move

WAKE UP YOUR SENSES

Push a marble softly across your table. Now push it harder. What was different?

♀ Let us find out

A thing that is still needs a push or a pull to start it moving. A **bigger push** sends it faster, and it travels further.

A push can **stop** a moving thing as well, or turn it a new way. When you catch a ball, your hands push it and it stops.

A ball rolls down a **ramp** with no push at all. The higher you lift the top of the ramp, the further the ball rolls.



A marble rolls down a ramp. A higher ramp sends it further.

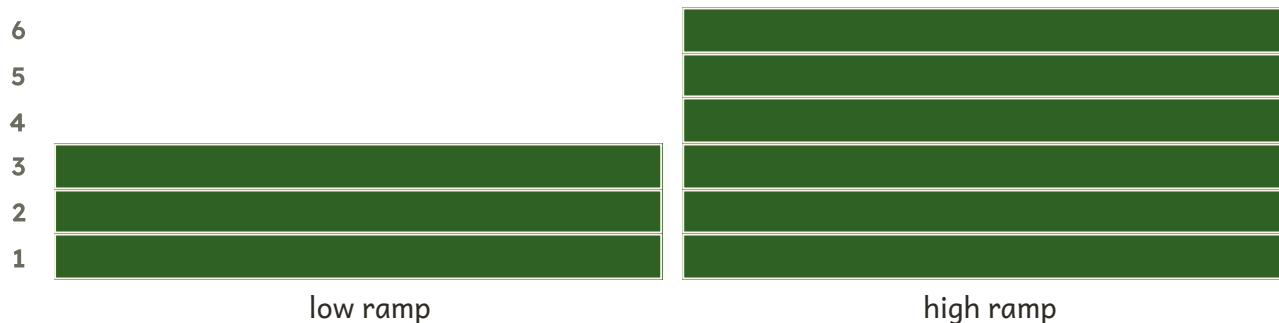
How do you know?

Dart wants his marble to roll right across the room. What should he do? How do you know?

Your turn**Watch me find out 3****Does a higher ramp send the marble further?**

- 1 We made a ramp with a tray resting on one book.
- 2 We rolled the marble down and counted hand spans.
- 3 Then we put the tray on three books and rolled it again.

We found out: the high ramp sent the marble **6** hand spans, and the low ramp sent it only **3**

Hand spans the marble rolled

- 1 Give the ball a small push. Say how far it rolled.
- 2 Now give it a big push. Which push sent it further?
- 3 Build a low ramp with a tray and one book.
- 4 Roll the marble down. Mark the spot where it stopped.
- 5 Lift the ramp on three books. Roll the marble again.
- 6 Which ramp sent the marble further, the low or the high one?

Your turn, and finding out together

Write how far it went each time.

what we did	how far the marble rolled
a small push	
a big push	
down the low ramp	
down the high ramp	

- 7 Put your four rolls in order from shortest to longest.
- 8 Draw the ramp you built. Show the marble rolling down.
- 9 How can you stop a rolling ball? Say two ways.
- 10 Catch a rolled ball. What did your hands do to it?
- 11 Why is a heavy log harder to move than a leaf?
- 12 Say one way to make a swing go higher.

The material you need

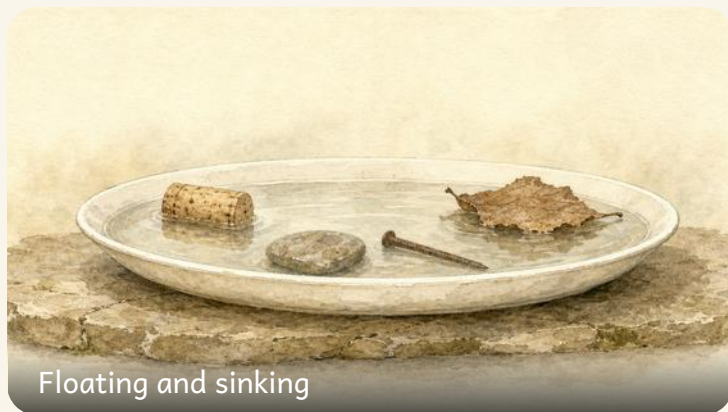
a smooth ball, a marble, a tray for the ramp, three books and some chalk to mark the floor

Find out together

Make a ramp with the tray and one book. Roll the marble down and chalk the spot where it stops. Now use three books. Roll it again and chalk that spot too. What changed?

Hedge bank challenge

Make your marble stop on a chalk cross on the floor. Change the height of your ramp until it lands there.



6.4 Floating and sinking

Do you remember?

You know that water can hold a boat up. What else have you seen sitting on top of water?

♀ Let us find out

Put something in water. Some things **float** and stay up on top. Some things **sink** and go down to the bottom.

The water gives a **push up** on everything you put into it.

Careful: it is not simply about heavy and light. An apple is heavy and it stays on top. A little nail goes straight down. So you must test each thing in water and see.



Some things stay up on top. Some go down to the bottom.

water

Draw each thing where it really went, on top or at the bottom.

How do you know?

Dart says the big apple must go to the bottom. Is Dart right? How do you know?

Your turn

WHAT I THINK WILL HAPPEN

Tick the things you think will float

WHAT REALLY HAPPENED

Draw or write what really happened

Test the apple and the nail. Then compare what you thought with what really happened.

 Watch me find out 4

Does the heavy apple go to the bottom?

- 1** Nib put the whole apple gently into the bowl of water.
 - 2** The apple bobbed up and sat on top of the water.
 - 3** Then she dropped in one small iron nail. Down it went.
-

We found out: the apple floats and the nail sinks, so a heavy thing does not always go to the bottom

- 1** Draw or write what you think will happen first.
- 2** Now test the apple in water. What really happened?
- 3** Compare what you thought with what really happened.
- 4** Name one thing that floats in your bath at home.
- 5** Draw the apple and the nail in the tank.
- 6** Say what the water does to a floating cork.

Your turn, and finding out together

Test each thing, then tick your chart.

thing	it floats	it sinks
cork	☐	☐
apple	☐	☐
candle	☐	☐
wooden peg	☐	☐
stone	☐	☐
coin	☐	☐
nail	☐	☐
marble	☐	☐
spoon	☐	☐
key	☐	☐

- 7 Tick it floats or it sinks for every thing above.
- 8 Put the coin in the water. What did it do?
- 9 Dart says anything heavy will sink. Is Dart right?
- 10 Order these four things from lightest to heaviest.
a cork, a wooden peg, an apple and a stone
- 11 Does the heaviest one in your row sink? Test it and say.
- 12 Find one more thing that floats. Draw it.

✂ The material you need

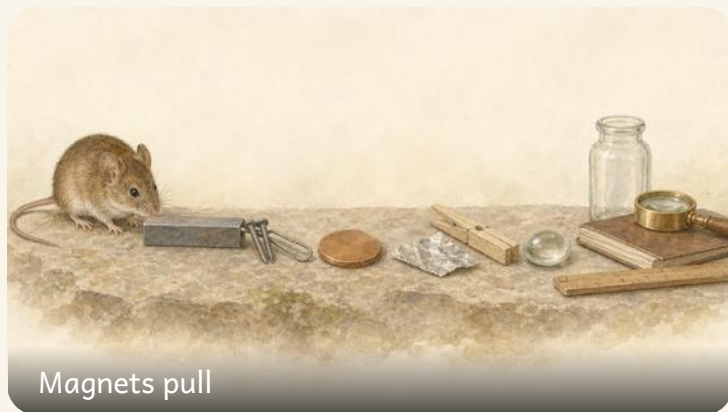
a bowl of water, a cork, an apple, a candle, a wooden peg, a coin, a nail, a marble, a spoon, a key, a stone and a towel

🔗 Find out together

Put each thing into the water on its own. Watch it, then tick your chart. When your chart is full, compare it with the chart your partner made. Are they the same?

! Take care

Wipe up any spilt water at once, so that nobody slips. Dry your hands before you write.



6.5 Magnets pull

WAKE UP
YOUR
SENSES

Hold a magnet near a paperclip. Move it closer, little by little. What happens?

♀ Let us find out

A **magnet** pulls some things towards it without even touching them. It pulls things made of **iron** and **steel**, like a steel paperclip or an iron nail.

A magnet does not pull wood, plastic, paper or glass.

Careful: a magnet does **not** pull every metal. A copper coin is metal, and the magnet leaves it alone. Aluminium foil is metal too, and the magnet leaves that alone as well. So test the metal, do not guess.



The magnet pulls the steel paperclip across the table.

How do you know?

Nib says a magnet pulls every metal. Is Nib right? How do you know?

Your turn

WHAT I THINK WILL HAPPEN

Tick the things you think the magnet will pull

WHAT REALLY HAPPENED

Draw or write what really happened

Test each thing with your magnet. Then compare your guess with what the magnet really did.

🔍 Watch me find out 5

Will this magnet pull the copper coin?

- 1 Dart held the magnet close to the steel paperclip.
- 2 The paperclip jumped across and stuck to the magnet.
- 3 Then he held the very same magnet close to the copper coin.
- 4 The coin did not move at all, and it is metal.

We found out: the magnet pulls the steel paperclip, but it does not pull the copper coin

1 Draw or write which things you think the magnet will pull.

2 Test each thing with the magnet. Tick your chart.

3 Compare your guess with what the magnet really did.

4 Name one thing that a magnet pulls.

5 Name one metal that a magnet does not pull.

6 Is a copper coin metal? Does the magnet pull it?

Your turn, and finding out together

thing	a magnet pulls it	a magnet does not pull it
steel paperclip	☐	☐
iron nail	☐	☐
copper coin	☐	☐
aluminium foil	☐	☐
wood	☐	☐
plastic	☐	☐



The magnet pulls the paperclip through a sheet of paper.

- 7** Hold the magnet near the paperclip. What happens?
- 8** Can the magnet pull the paperclip through paper? Test it.
- 9** Sort the things into two trays.
One tray for things the magnet pulls, one tray for all the rest.
- 10** Draw the magnet pulling the nail. Add an arrow for the pull.
- 11** Nib says a magnet pulls every metal. Say why Nib is wrong.
- 12** Say two things you found out about magnets today.

✂ The material you need

one magnet, a steel paperclip, an iron nail, a copper coin, some aluminium foil, a wooden peg and a sheet of paper

♂ Find out together

Hold the magnet under the paper. Put the paperclip on top of the paper. Move the magnet slowly. Does the paperclip follow it? Now try the copper coin, and compare what happens.

! Take care

Keep magnets well away from screens, clocks and cards with a stripe on the back.

Words we used in Unit 6

▣ WORDS WE USED



push

you move something away from you



pull

you move something towards you



force

a push or a pull



roll

to move along by turning over and over



float

it stays up on top of the water



sink

it goes down under the water



magnet

it pulls some metals, but not all



iron

a metal that a magnet does pull

☐ Check what you can do

- 1** Say the difference between a push and a pull.
- 2** Name three ways a thing can move.
- 3** Say what makes a ball roll faster down a ramp.
- 4** Name one heavy thing that floats.
- 5** Explain why a magnet picks up a nail but not a copper coin.

Look what I can do!

- ☐ I can tell a push from a pull.
- ☐ I can name different ways things move.
- ☐ I can make something move faster or slower.
- ☐ I can say which things float and which sink.
- ☐ I know a magnet pulls some metals but not all of them.
- ☐ I can say what I found out, and how I know.

	confident	almost there	needs more work
telling a push from a pull	☐	☐	☐
making things move	☐	☐	☐
testing what floats	☐	☐	☐
testing what a magnet pulls	☐	☐	☐
saying how I know	☐	☐	☐



Dart, mid jump, with everything the class tested this year.

For teachers

Two misconceptions are worth naming out loud in this unit. The first is that heavy things sink: the apple and the nail settle it in one minute, and a ship is the argument that finishes it. The second is that magnets pull all metals: the copper coin and the aluminium foil are in the kit for exactly that reason, so do not leave them out.

This is the last unit, so it is the right place to look back. Ask pupils what they believed in September that they now know is wrong. Being able to answer that is the single best sign of a year of real science.

Unit 1 project • Our class field guide

ALONG THE WAY, UNIT 1

■ Our class field guide

Make one page for the class field guide. Your page shows one real plant you found and pressed.

✂ The material you need

one leaf or one whole small weed, old newspaper, two heavy books, a sheet of card, glue and a pencil

- 1 Pick one leaf from the ground. Do not pull up a rare plant.
- 2 Lay the leaf flat between two sheets of newspaper.
- 3 Put the heavy books on top. Leave it for four days.
- 4 Glue your dry, flat leaf onto the card.
- 5 Write its name, and where you found it.
- 6 Put every page together to make the class field guide.

• CONTINUED



A pressed leaf glued to card, named in the finder's own hand.

What did you find out?

Which part of the plant did you press? Say how you know it was alive.

**Take care**

Wash your hands after handling plants and soil.

Unit 2 project · Our own shaker

ALONG THE WAY, UNIT 2

■ Our own shaker

Make a shaker that makes a sound. Then show on a chart when it is loud and when it is quiet.

✂ The material you need

a clean empty tub with a lid, dry rice, dry pasta, sticky tape and a strip of paper for your chart

- 1 Put one spoon of rice into the tub. Press the lid on.
- 2 Tape the lid down so that nothing spills out.
- 3 Shake it gently. Listen to the quiet sound.
- 4 Now shake it hard. Listen to the loud sound.
- 5 Draw a small bar for quiet and a tall bar for loud.
- 6 Swap the rice for pasta, and listen again.

• CONTINUED



A taped tub of rice beside a chart of quiet and loud bars.

What did you find out?

Which shaking made the loudest sound? Say how you know.

Unit 3 project · Four trays of materials

ALONG THE WAY, UNIT 3

■ Four trays of materials

Sort a tray of things by what they are made of. Make four labelled trays for the class table.

✂ The material you need

a box of safe classroom things, four shallow trays, four cards and a pencil

- 1 Write one card for each tray: wood, metal, plastic, cloth.
- 2 Take one thing out of the box. Feel it and look at it.
- 3 Say what it is made of. Put it in the right tray.
- 4 Do the same with every other thing in the box.
- 5 Count the things in each tray. Write the number on the card.
- 6 Show your trays to another group and talk about them.

• CONTINUED



Four trays of sorted things, each with its own written card.

What did you find out?

Which tray held the most things? Say how you sorted them.

Unit 4 project · Our layered soil jar

ALONG THE WAY, UNIT 4

■ Our layered soil jar

Make a soil jar and watch the soil settle. You will see the layers inside your soil.

✂ The material you need

a clear jar with a lid, two spoons of garden soil, water and a marker pen

- 1 Spoon the soil into the jar. Fill the jar with water.
- 2 Screw the lid on tightly. Check that it cannot leak.
- 3 Shake the jar hard for ten seconds. The water goes cloudy.
- 4 Stand the jar on the table. Do not touch it again.
- 5 Look at it after one hour, and then the next day.
- 6 Mark each layer you can see with the pen.

• CONTINUED



A jar of shaken soil settling into layers of grit and mud.

What did you find out?

Which bits sank to the bottom first? Say how you know.

**Take care**

Keep the lid screwed on. Wash your hands after touching soil.

Unit 5 project • The five senses game

ALONG THE WAY, UNIT 5

■ The five senses game

Make a guessing game for a friend. They use one sense at a time to name each thing.

✂ The material you need

a cloth bag, five safe things from the classroom, a scarf for a blindfold and a card for scores

- 1 Put five safe things into the bag. Keep them a secret.
- 2 Ask your friend to reach in and feel one thing.
- 3 Can they name it by touch alone? Write yes or no.
- 4 Now let them shake it and listen to it.
- 5 Last of all, let them look at it.
- 6 Swap over, and play the game again.

• CONTINUED



A cloth bag of hidden things beside a card of yes and no ticks.

What did you find out?

Which sense helped your friend most? Say how you know.

**Take care**

Never taste anything in the bag. Only feel, listen and look.

Unit 6 project · Our ramp test

ALONG THE WAY, UNIT 6

■ Our ramp test

Test three things on one ramp. Find out which one rolls the furthest of the three.

✂ The material you need

a smooth tray for the ramp, three books, a ball, a marble, a wooden peg and chalk

- 1 Rest one end of the tray on the three books.
- 2 Let the ball go at the top. Do not push it.
- 3 Chalk the floor where the ball stopped rolling.
- 4 Do the same with the marble, then the wooden peg.
- 5 Measure each chalk mark in hand spans. Write the numbers down.
- 6 Put your three things in order from shortest roll to longest.

• CONTINUED



A tray ramp with chalk marks showing how far each thing rolled.

What did you find out?

Which thing rolled the furthest? Say how you made the test fair.

Our own field guide



Every page the class made this year, gathered into one book.

THE LAST PROJECT

■ Make a class field guide

All year you have drawn what you found on the hedge bank. Now you will put every page together and make one book for your class.

It will be a **field guide**: a book you can carry outside and look things up in.

✂ The material you need

every page your class has made this year, a hole punch, string or treasury tags, and two sheets of thick card for the covers

How to make your field guide

- 1 Spread all your pages on the floor. Look at what you have.
- 2 Sort them into groups. Living things in one pile, rocks in another.
- 3 Put each group in an order that makes sense to you.
- 4 Design a cover. Write the title and your class name on it.
- 5 Punch holes down one side of every page.
- 6 Thread the string through, and tie it firmly.

1 What did you call your field guide?

2 How did you sort your pages? Say your rule.

3 Which page are you proudest of? Say why.

4 What did you find out this year that surprised you most?

Hedge bank challenge

Swap field guides with another class. Can you find something in theirs that is not in yours?

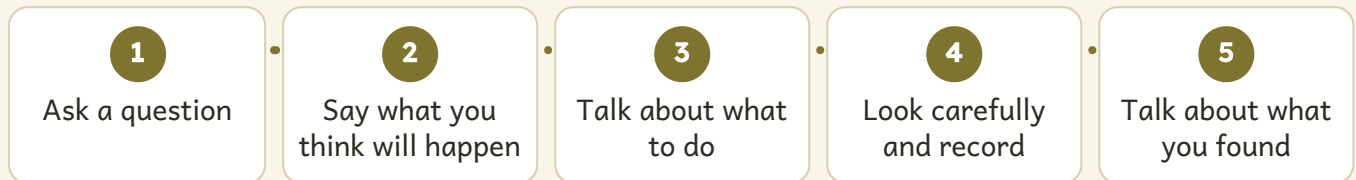
Take care

Take care with the hole punch, and ask an adult to help you.

What a scientist does

You have done real science this year. Here are the five steps you used, every single time.

The five steps, one more time



You will use these same five steps next year, and the year after that. Every scientist in the world uses them.



A year of finding out, from the first bean shoot to the last ramp test.

Look what I can do now!

- ☐ I can ask a question I could really find out about.
- ☐ I can say what I think will happen, before I start.
- ☐ I can look carefully, and use a hand lens.
- ☐ I can write down or draw what I saw.
- ☐ I can say what I found out, and say how I know.
- ☐ I can say when what happened surprised me.

	confident	almost there	needs more work
asking a good question	☐	☐	☐
saying what I thought would happen	☐	☐	☐
looking closely	☐	☐	☐
writing down what I saw	☐	☐	☐
saying what I found out	☐	☐	☐

For teachers

These six statements map onto the Thinking and Working Scientifically strand at Stage 1. Pupils who can do the last one, saying when a result surprised them, are doing the most difficult and most valuable part of the strand, and it is worth naming that out loud.

The five-step chart appears three times in this book: in the front matter, here, and in the teacher note. All three are generated from one list, so if you adapt the wording for your class it stays consistent.

WORD LIST

Every word we learnt

Every science word in this book, in the order of the alphabet. The name beside each one tells you where you met it.

A

alive it grows, feeds and makes
youngUnit 1

B

body all of you, from head to
toeUnit 5

C

clay soft sticky earth that you
can shapeUnit 3

D

different not the same as
another oneUnit 5

dish for water, soil or
seedsGetting set up

E

WORD LIST · CONTINUED

ear the part we hear withUnit 2
Earth the planet we all live onUnit 4

F

far away a long way from youUnit 2
float it stays up on top of the waterUnit 6

elastic it springs back to its own shapeUnit 3

flower makes the seedsUnit 1
force a push or a pullUnit 6

G

glass hard, smooth and see-throughUnit 3

H

hand lens makes small things look biggerGetting set up
healthy well, and looked afterUnit 5

hearing what we find out with our earsUnit 5
hollow empty inside, so it can ringUnit 2

I

iron a metal that a magnet does pullUnit 6

J

jar for holding what you findGetting set up

WORD LIST

Every word we learnt, continued

L

leaf makes food out of lightUnit 1 **loud** a big soundUnit 2

listen to use your ears on
purposeUnit 2

M

magnet for finding out what it
pullsGetting set up

material the stuff a thing is
made ofUnit 3

metal hard and often shinyUnit 3

N

WORD LIST · CONTINUED

never alive it has never grown or
fedUnit 1

notebook for drawing what you
seeGetting set up

P

pencil for writing and
drawingGetting set up

plastic made by people, keeps
water outUnit 3

pull you move something towards
youUnit 6

push you move something away
from youUnit 6

Q

quiet a small soundUnit 2

R

rock hard stone from the
groundUnit 4

roll to move along by turning
over and overUnit 6

root holds the plant, drinks
waterUnit 1

ruler for measuring how
longGetting set up

WORD LIST

Every word we learnt, continued

S

sand tiny loose grains of worn rockUnit 4

seed a new plant grows from itUnit 1

senses the five ways we find things outUnit 5

shadow made where something blocks the lightUnit 4

sink it goes down under the waterUnit 6

smell what we find out with our noseUnit 5

soil the loose dark earth plants grow inUnit 4

sound what we hear with our earsUnit 2

stem holds the plant upUnit 1

WORD LIST · CONTINUED

T

taste what we find out with our tongueUnit 5

the Sun it gives us heat and lightUnit 4

touch what we feel with our skinUnit 5

trowel for digging a littleGetting set up

V

vibrate to shake to and fro very fastUnit 2

W

wood it comes from a tree and has grainUnit 3

wool soft and warm, from a sheepUnit 3

worm it mixes the soil as it movesUnit 4

About this book

Science & Lab, Year 1 supports the Cambridge Primary Science curriculum framework at Stage 1, across the Biology, Chemistry, Physics and Earth and Space strands, with Thinking and Working Scientifically running through every topic rather than sitting apart from them.

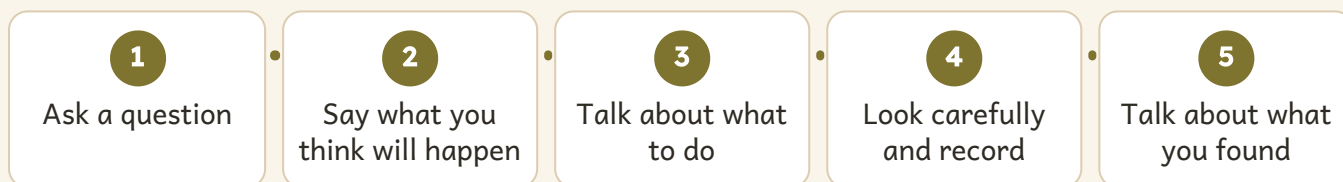
How the book is built. Six units, twenty-four topics. Each topic runs to three pages: one teaching page, then two pages of practice. Every topic carries at least twelve practice items, one modelled enquiry with a stated finding, and at least one *How do you know?* prompt. Every unit ends with a word list, a review, and a page for pupils to record what they can now do.

What the enquiry steps are for. The five steps on page 11 recur in every unit and again on the closing pages. They are deliberately the same five words each time, so that by the summer term a pupil can name the step they are on without prompting.

Safety. Practical work is written for five and six year olds under direct supervision. Carry out your own risk assessment first, and follow your school policy. The Sun rule in Unit 4 is the one to repeat aloud at every outdoor session rather than reading once.

Misconceptions this book tackles head on. That plants eat soil (Unit 1, revisited in Unit 4). That sound travels through nothing (Unit 2). That heavy things always sink (Unit 6). That magnets pull every metal (Unit 6, where a copper coin and aluminium foil are in the kit for that purpose). In each case the wrong idea is named, put in a character's mouth, and then tested, because a misconception that is never spoken aloud is never given up.

The five enquiry steps, as pupils meet them



For teachers

The five steps above are generated from one list shared with page 11 and the closing skills pages, so all three always match.

Assessment is built in rather than bolted on. The review questions at the end of each unit are pitched to be answered aloud, and the can-do page beside them is for the pupil to complete, not the teacher. A pupil who ticks a statement they cannot yet demonstrate has told you something useful.

PRIME BOOKS

Science

Student Manual

How do you know?

Come out to the hedge bank, where a badger with a brass hand lens and a wood mouse with enormous ears want to know how everything works.

INSIDE THIS BOOK

- Twenty-four topics across six units
- Over 280 things to do, say and draw
- A modelled enquiry in every single topic
- Six projects, and a class field guide to finish
- A word list of every science word you meet

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